



Workshop Manual

Crafter 2017 ➤

TGE 2017 ➤

8-speed automatic gearbox 0DR

Edition 09.2018



List of Workshop Manual Repair Groups

Repair Group

- 00 - Technical data
- 32 - Torque converter
- 37 - Controls, housing
- 38 - Gears, control

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.



Contents

00 - Technical data	1
1 Safety information	1
1.1 Rules for safety when working on the gearbox	1
1.2 Safety measures when working on vehicles with a start/stop system	2
1.3 Safety precautions when using testers and measuring instruments during a road test	2
1.4 Safety precautions when tow-starting or towing	2
2 Repair notes	3
2.1 Rules for cleanliness	3
2.2 General notes	3
2.3 General repair instructions	3
2.4 Contact corrosion	4
2.5 Gaskets and seals	5
2.6 Nuts and bolts	5
2.7 Line routing and attachment	5
3 Technical data	6
3.1 Capacities	6
3.2 Allocation gearbox - engine	6
4 Identification	7
4.1 Gearbox identification	7
4.2 Identification of torque converter	7
5 Electrical components	8
5.1 Overview of fitting locations - electrical components	8
32 - Torque converter	9
1 Torque converter	9
1.1 Assembly overview - torque converter	9
1.2 Removing and installing torque converter	9
1.3 Draining torque converter	11
1.4 Removing and installing torque converter oil seal	11
37 - Controls, housing	13
1 ATF circuit	13
1.1 Assembly overview - ATF circuit	13
1.2 Removing and installing ATF cooler	14
1.3 Removing and installing ATF filter	15
1.4 Removing and installing ATF oil pressure lines	16
2 ATF	20
2.1 Checking ATF level	20
2.2 Draining and filling ATF	22
2.3 Purging ATF circuit	26
3 Removing and installing gearbox	30
3.1 Specified torques for gearbox	30
3.2 Removing gearbox	30
3.3 Installing gearbox	34
4 Assembly mountings	37
4.1 Assembly overview - assembly mountings	37
4.2 Assembly overview - gearbox carrier with gearbox mounting	37
4.3 Removing and installing gearbox mounting	38
4.4 Removing and installing gearbox carrier	38
5 Transporting gearbox	40
6 Securing to engine and gearbox support	42



7	Selector mechanism	43
7.1	Assembly overview - selector mechanism	43
7.2	Checking selector mechanism	44
7.3	Emergency release from position P	45
7.4	Removing and installing selector mechanism	45
7.5	Removing and installing selector lever handle	47
7.6	Removing and installing gear selector cable	48
7.7	Checking and adjusting selector lever cable	50
38	Gears, control	51
1	ATF system	51
1.1	Assembly overview - sump/oil strainer	51
1.2	Removing and installing sump	52
1.3	Removing and installing hydraulic impulse oil storage with accumulator solenoid N485 ..	54
2	Oil seals in gearbox	56
2.1	Assembly overview - output shaft flange	56
2.2	Renewing output shaft seal	56
2.3	Removing and installing output shaft flange	58
3	Valve body	60
3.1	Removing and installing valve body	60
4	Gearbox control system	63
4.1	Removing and installing automatic gearbox control unit J217	63



00 – Technical data

1 Safety information

(VRL012223; Edition 09.2018)

⇒ [“1.1 Rules for safety when working on the gearbox”, page 1](#)

⇒ [“1.2 Safety measures when working on vehicles with a start/stop system”, page 2](#)

⇒ [“1.3 Safety precautions when using testers and measuring instruments during a road test”, page 2](#)

⇒ [“1.4 Safety precautions when tow-starting or towing”, page 2](#)

1.1 Rules for safety when working on the gearbox

To prevent personal injury and material damage to the vehicle, observe the following:



WARNING

Risk of injury and accident by accidental engagement of gear while engine is running.

- ◆ *Before working on vehicle with engine running, always move selector lever into position “P” and apply handbrake.*

Risk of poisoning from exhaust gases while engine is running.

- ◆ *While the engine is running, an exhaust gas extraction system must be attached to the exhaust system.*



WARNING

Risk of injury from radiator fan starting automatically.

- ◆ *Disconnect electrical connectors before starting work in the area of the radiator cowl.*

To prevent personal injury and damage to or destruction of electrical and electronic components, observe the following:

- ◆ Connect and disconnect measuring and testing devices only with the ignition switched off.



Caution

Danger of destruction of electronic components when battery is disconnected.

- ◆ *Observe the necessary measures when disconnecting the battery.*
- ◆ *Disconnect battery only after the ignition has been switched off.*



1.2 Safety measures when working on vehicles with a start/stop system

Risk of injury due to unexpected motor start

If the vehicle's start/stop system is activated, the engine can start unexpectedly. A message in the dash panel insert indicates whether the Start/Stop system is activated.

- Deactivate Start/Stop system by switching off the ignition.

1.3 Safety precautions when using testers and measuring instruments during a road test

Risk of injury from unsecured test instruments and measuring equipment

When the front passenger airbag is triggered in an accident, insufficiently secured testing and measuring instruments become dangerous projectiles.

- Secure testing and measuring instruments on the rear seat.

Or

- Have a second person operate the test instruments and measuring equipment on the rear seat.

1.4 Safety precautions when tow-starting or towing



Caution

Danger of irreparable damage to gearbox.

- ◆ *Selector lever must be in "N" position for towing vehicle.
Do not tow for more than 50 km or faster than 50 km/h.*



Note

It is not possible to tow-start the motor.



2 Repair notes

⇒ [“2.1 Rules for cleanliness”, page 3](#)

⇒ [“2.2 General notes”, page 3](#)

⇒ [“2.3 General repair instructions”, page 3](#)

⇒ [“2.4 Contact corrosion”, page 4](#)

⇒ [“2.5 Gaskets and seals”, page 5](#)

⇒ [“2.6 Nuts and bolts”, page 5](#)

⇒ [“2.7 Line routing and attachment”, page 5](#)

2.1 Rules for cleanliness

- ◆ Thoroughly clean all joints and surrounding areas before dismantling.
- ◆ Use commercially available, lint-free cloths for cleaning.
- ◆ Seal open lines and connections immediately using clean plugs or sealing caps from engine bung set - VAS 6122- .
- ◆ Place removed parts on a clean surface, and cover them over. Use plastic foils or lint-free cloths.
- ◆ If repair work cannot be performed immediately, cover or seal new parts which have been removed from their packing.
- ◆ Install only clean parts; do not remove new parts from packaging until immediately before installing.
- ◆ Protect separated electrical connectors from dirt and water and only reconnect in dry condition.

2.2 General notes

- ◆ The engine control unit has a self-diagnosis capability. Before carrying out repairs and for fault finding, first read event memory.
- ◆ For trouble-free operation of electrical components, a voltage of at least 11.5 volts is necessary.
- ◆ Do not use sealants containing silicone. Particles of silicone drawn into the engine will not be burnt in the engine and damage the lambda probe.
- ◆ Vehicles are fitted with a crash fuel shut-off circuit. It reduces the danger of a fire in a crash as the fuel pump is switched off via the fuel pump relay.
- ◆ The system also improves the starting characteristics of the engine. When the driver door is opened, the fuel pump is activated for 2 seconds to develop pressure in the fuel system.

2.3 General repair instructions

To ensure flawless and successful gearbox repairs, the greatest care and cleanliness as well as the use of good and proper tools are essential. Of course, the basic rules for safety also apply during all repair work.

A number of general notes on the individual repair procedures, which can otherwise be found in the relevant sections of the workshop manual, are summarised here. They apply for this particular workshop manual.



Notes on the gearbox

- ◆ If gearbox covers have been unbolted or gearbox has no fluid, do not run engine or tow vehicle either.
- ◆ First thoroughly clean connecting points and surrounding areas and then loosen bolts.
- ◆ When installing the gearbox, ensure that dowel sleeves between the engine and gearbox are correctly located.
- ◆ Place removed parts on a clean surface and cover them to prevent them from getting dirty. Use sheeting and paper for this purpose. Use only lint-free cloths.
- ◆ If repair work cannot be performed immediately, carefully cover or seal components.
- ◆ Install only clean parts; do not remove new parts from packaging until immediately before installing.
- ◆ Do not run engine or tow vehicle with pan removed or when there is no ATF in gearbox.
- ◆ With gearbox removed, secure torque converter against falling out.
- ◆ After installing, check ATF level ⇒ [page 20](#) .

Sealant

- ◆ Thoroughly clean housing joint surfaces before applying sealant.
- ◆ Evenly apply a thin coat of sealant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis for MAN TGE.
- ◆ Breather holes must remain free of sealant.

2.4 Contact corrosion

Contact corrosion can occur if unsuitable fastening elements (screws, bolts, nuts, washers) are used.

For this reason, only connecting elements with a special surface coating have been fitted.

In addition, rubber, plastic and adhesives are made of non-conductive materials.

If there are doubts about the suitability of fastening components, always use new parts ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis for MAN TGE.

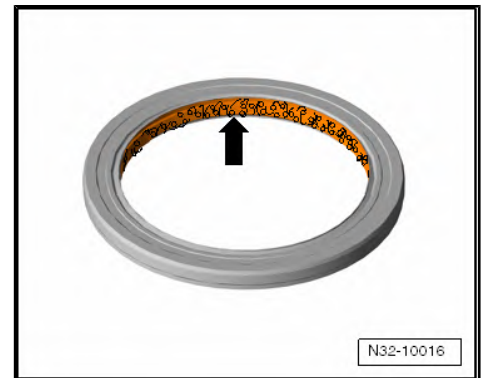
Observe the following:

- ◆ It is recommended to use only genuine parts, since only genuine parts are approved.
- ◆ Damage resulting from contact corrosion is not covered by the warranty.



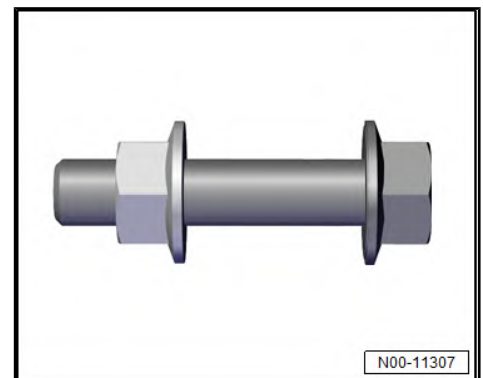
2.5 Gaskets and seals

- ◆ Renew O-ring, seals and gaskets.
- ◆ After removing gaskets and seals, always inspect contact surface of housings or shafts for burrs or other signs of damage.
- ◆ Before installing the seal, slightly moisten outer circumference of seal and sealing lip with ATF depending on installation location.
- ◆ Only use ATF for automatic transmission components. Other types of lubrication cause faults to occur in the gearbox hydraulics.
- ◆ Open-side of seals point towards oil.
- ◆ After installing, check ATF level ⇒ [page 20](#) .



2.6 Nuts and bolts

- ◆ Loosen and tighten securing bolts and nuts for covers and housings diagonally.
- ◆ Specified torques given are for uncoiled nuts, bolts and screws.
- ◆ Clean threads of bolts which are to be installed with locking fluid using a wire brush.
- ◆ Use a thread chaser to clear residual locking fluid from all threaded holes into which self-locking bolts are to be screwed. Otherwise there is a danger of bolts shearing when subsequently being removed.
- ◆ Always renew self-locking bolts and nuts.
- ◆ Specified torques given are for uncoiled nuts, bolts and screws.



2.7 Line routing and attachment

Risk of damage to lines

Lines may become damaged by moving or hot components.

- Route lines in their original positions.
- Ensure there is sufficient clearance to moving or hot components.



3 Technical data

⇒ ["3.1 Capacities", page 6](#)

⇒ ["3.2 Allocation gearbox - engine", page 6](#)

3.1 Capacities



Caution

Risk of malfunctions or failure of gearbox.

- ◆ *Only gear oil available as a replacement part may be used for the manual gearbox 0DR ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis for MAN TGE.*

Capacities	
Recharging the gearbox	7.8 l

For allocation of ATF, refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

3.2 Allocation gearbox - engine

Gearbox		8-speed automatic gearbox 0DR
Gearbox	Codes	RVN, RVP
Allocation	Model	Crafter 2017 ►, TGE 2017 ►
	Engine	2.0 l - 120 kW bi-turbo diesel
Ratio	1st gear	4.714
	2nd gear	3.143
	3rd gear	2.106
	4th gear	1.667
	5th gear	1.285
	6th gear	1
	7th gear	0.839
	8th gear	0.667
	Reverse gear	3.317

Refer to ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis (MAN TGE) for the following data:

- ◆ ATF specifications
- ◆ Specification for gear oil for rear final drive
- ◆ Rear final drive is allocated based on engine code



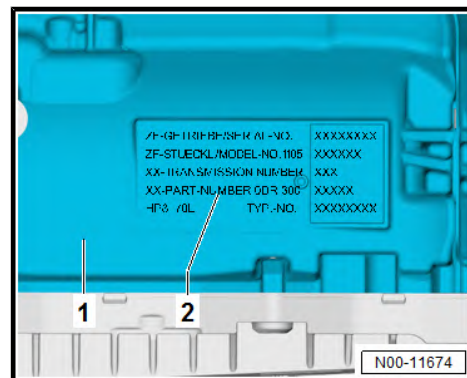
4 Identification

⇒ [“4.1 Gearbox identification”, page 7](#)

⇒ [“4.2 Identification of torque converter”, page 7](#)

4.1 Gearbox identification

Gearbox -1- identification and data -2-



4.2 Identification of torque converter

There are various types of torque converter. They are identified by codes.

Allocation ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis for MAN TGE.



5 Electrical components

⇒ "5.1 Overview of fitting locations - electrical components", page 8

5.1 Overview of fitting locations - electrical components

1 - Valve body

- ❑ Fitting location
⇒ [Item 1 \(page 51\)](#)
- ❑ Removing and installing
⇒ [page 60](#)

2 - Automatic gearbox control unit - J217-

- ❑ Removing and installing
⇒ [page 63](#)

3 - Engine control unit - J623-

- ❑ Removing and installing
⇒ Rep. gr. 24 ; Engine control unit; Removing and installing engine control unit - J623-

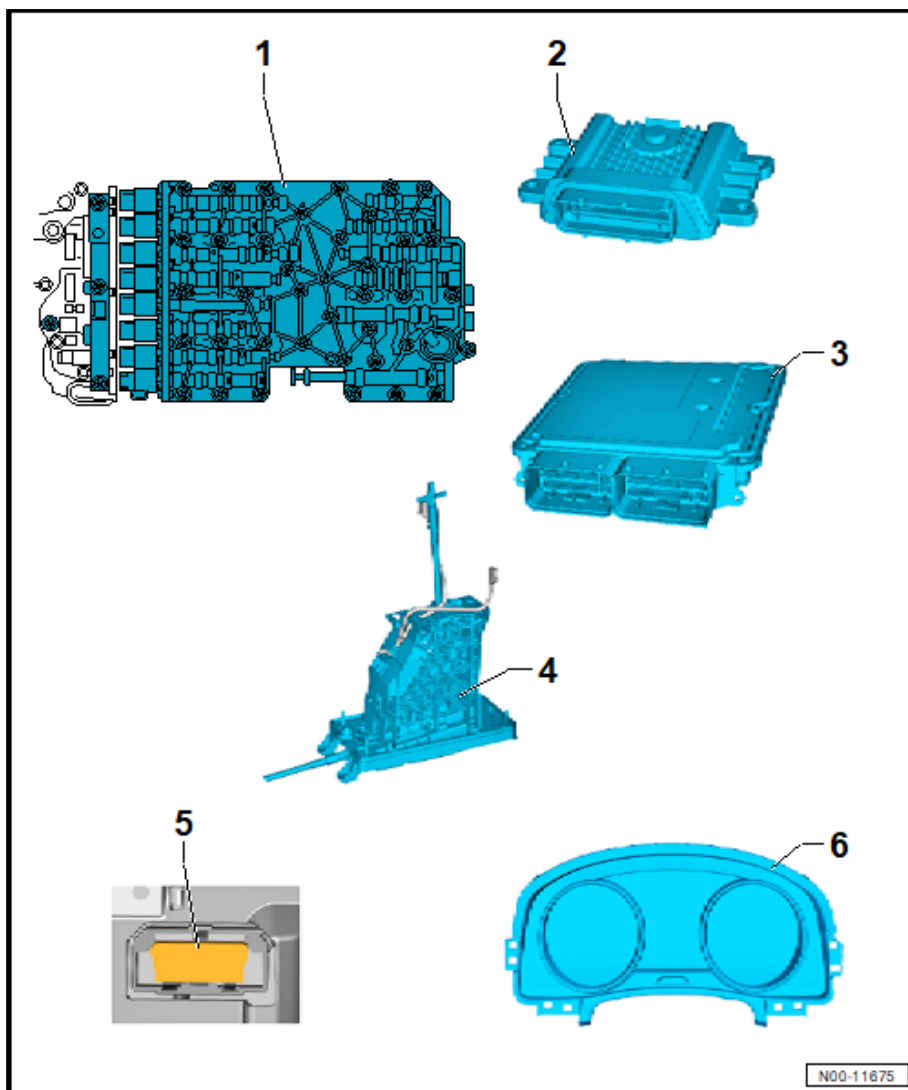
4 - Selector mechanism

- ❑ With Tiptronic switch - F189-
- ❑ Removing and installing
⇒ [page 45](#)

5 - Diagnostic connection

6 - Selector lever position indicator - Y6-

- ❑ Removing and installing
⇒ Electrical system; Rep. gr. 90 ; Dash panel insert; Assembly overview - dash panel insert





32 – Torque converter

1 Torque converter

⇒ [“1.1 Assembly overview - torque converter”, page 9](#)

⇒ [“1.2 Removing and installing torque converter”, page 9](#)

⇒ [“1.3 Draining torque converter”, page 11](#)

⇒ [“1.4 Removing and installing torque converter oil seal”, page 11](#)

1.1 Assembly overview - torque converter

1 - Automatic gearbox

- ☐ Removing ⇒ [page 30](#)
- ☐ Installing ⇒ [page 34](#)

2 - Seal

- ☐ Removing and installing
⇒ [page 11](#)

3 - Seal

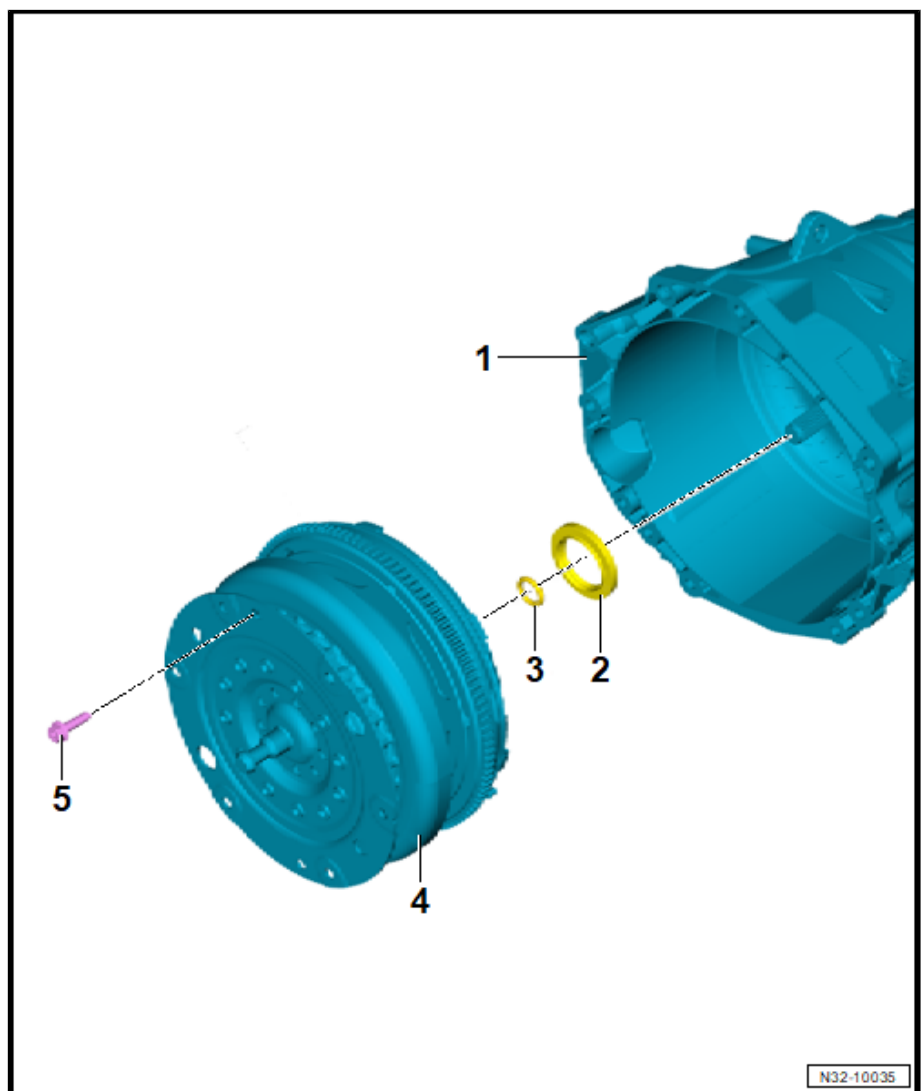
- ☐ Renew after removal

4 - Torque converter

- ☐ Removing and installing
⇒ [page 9](#)

5 - Bolts

- ☐ Renew after removal
- ☐ Qty. 3
- ☐ 85 Nm



1.2 Removing and installing torque converter

Special tools and workshop equipment required



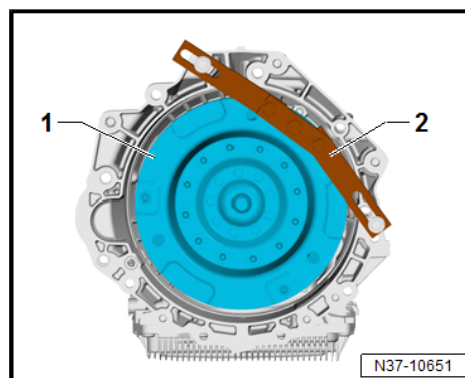
- ◆ Drip tray for workshop hoist - VAS 6208-



- ◆ Turbocharger tester - V.A.G 1397B-

Removing

- Remove gearbox ➔ [page 30](#) .
- If present, remove support bridge - 30-211A- -2- from torque converter -1-.
- Place drip tray for workshop hoist - VAS 6208- under gearbox.



- Remove torque converter in direction of -arrow- from gearbox.

Note

Oil may escape when the torque converter is pulled out of the gearbox.

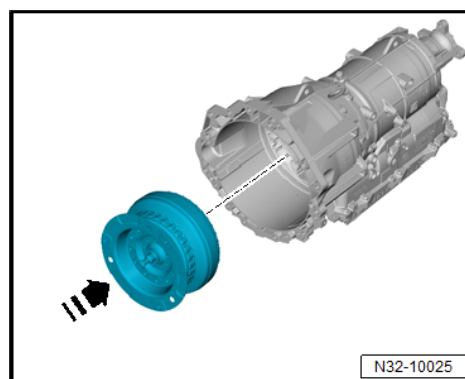
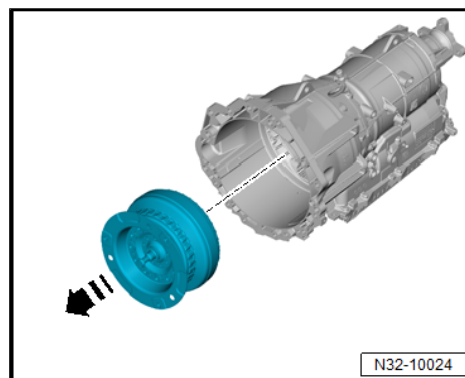
Installing

Install in reverse order of removal, observing the following:

- Push torque converter in direction of -arrow- into gearbox as far as stop.

Note

- The torque converter is properly installed if it can be turned easily by hand and sits around the whole circumference at an equal depth in the gearbox.

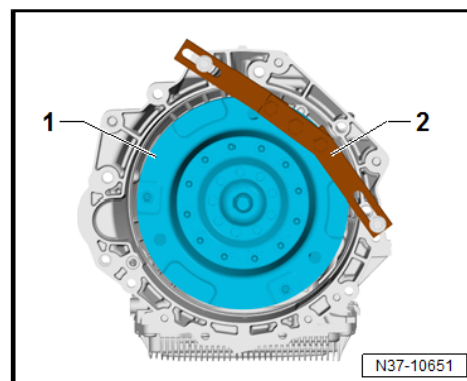




Note

If the gearbox is not installed again immediately, secure the torque converter.

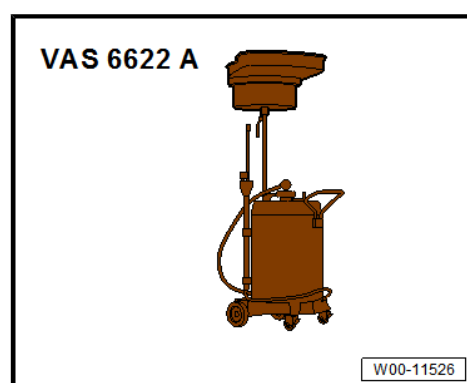
- Secure torque converter -1- with support bridge - 30-211A- -2-.



1.3 Draining torque converter

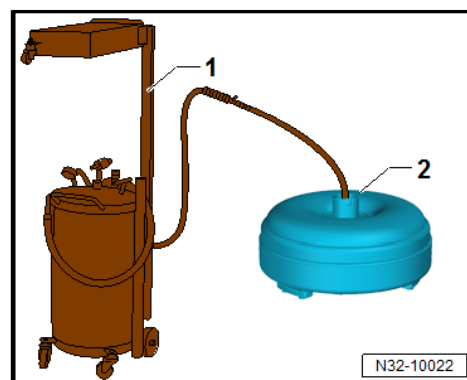
Special tools and workshop equipment required

- ◆ Used oil collector and extractor - VAS 6622A-



Sequence of operations

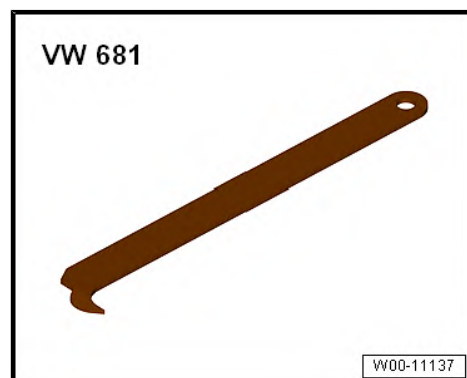
- Extract ATF from torque converter -2- using used oil collection and extraction unit - VAS 6622A- -1-.



1.4 Removing and installing torque converter oil seal

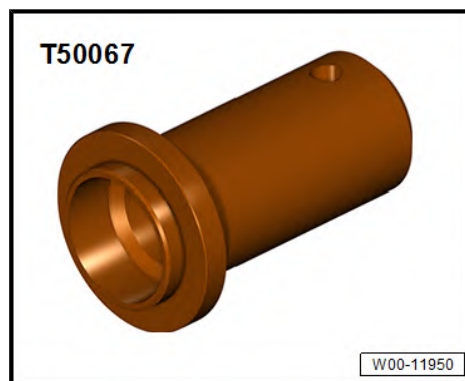
Special tools and workshop equipment required

- ◆ Removal tool - VW 681-





◆ Thrust piece - T50067-



◆ Depth gauge

Removing

- Remove torque converter ➔ [page 9](#) .
- Calculate installation depth of seal using depth gauge and note result.
- Pry out seal with lever - VW681- -1-.

Installing

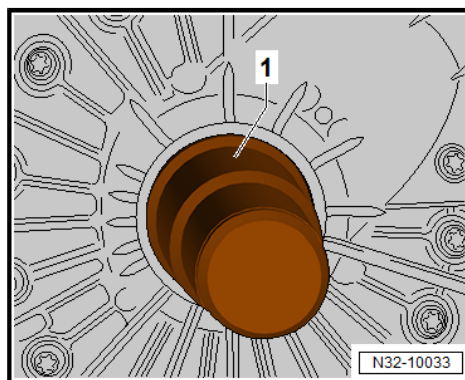
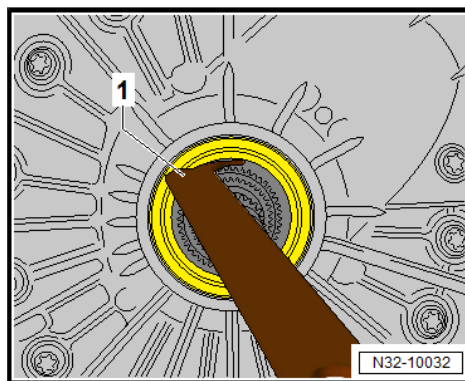
Install in reverse order of removal, observing the following:

- Apply ATF to circumference and sealing lips of oil seal.

Installation position:

Open side of oil seal faces gearbox.

- Press oil seal in straight as far as possible by hand.
- Note installation depth of seal.
- Drive in seal with thrust piece - T50067- -1-.
- Install torque converter ➔ [page 9](#) .





37 – Controls, housing

1 ATF circuit

⇒ [“1.1 Assembly overview - ATF circuit”, page 13](#)

⇒ [“1.2 Removing and installing ATF cooler”, page 14](#)

⇒ [“1.3 Removing and installing ATF filter”, page 15](#)

⇒ [“1.4 Removing and installing ATF oil pressure lines”, page 16](#)

⇒ [“1.4.1 Removing and installing oil temperature regulator”, page 18](#)

1.1 Assembly overview - ATF circuit

1 - Gearbox

- ☐ Removing ⇒ [page 30](#)
- ☐ Installing ⇒ [page 34](#)

2 - Seals

- ☐ Renew after removal
- ☐ Qty. 8

3 - Bracket

- ☐ Qty. 2

4 - Bolts

- ☐ Qty. 3
- ☐ 8 Nm

5 - Bolt

- ☐ Renew after removal
- ☐ 8 Nm

6 - Oil pressure lines

- ☐ Qty. 2
- ☐ Removing and installing
⇒ [page 16](#)

7 - Bolt

- ☐ Renew after removal
- ☐ 8 Nm

8 - Oil temperature regulator

- ☐ Removing and installing
⇒ [page 18](#)

9 - Bolts

- ☐ Renew after removal
- ☐ Qty. 3
- ☐ 8 Nm

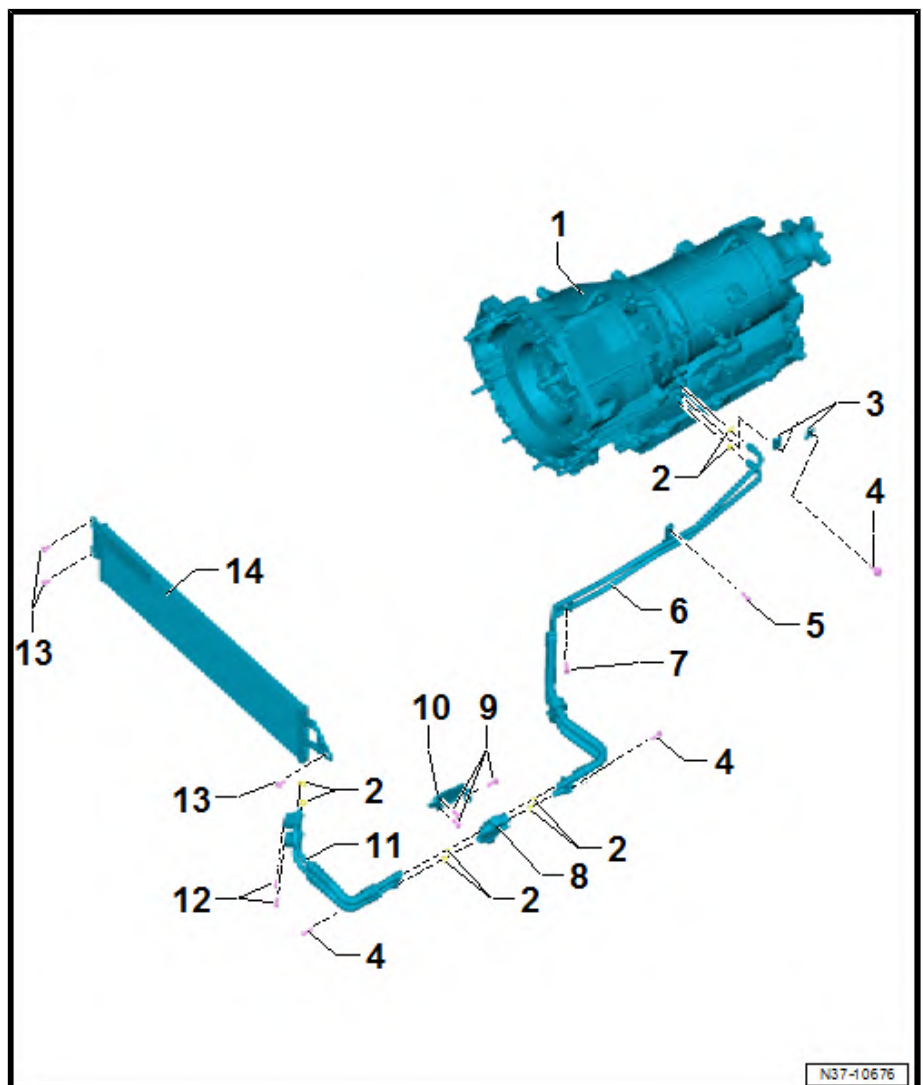
10 - Bracket

11 - Oil pressure lines

- ☐ Removing and installing ⇒ [page 16](#)

12 - Bolts

- ☐ Qty. 2





- ☐ 8 Nm

13 - Bolts

- ☐ Qty. 3
- ☐ 2 Nm

14 - ATF cooler

- ☐ Removing and installing ⇒ [page 14](#)

1.2 Removing and installing ATF cooler

Special tools and workshop equipment required

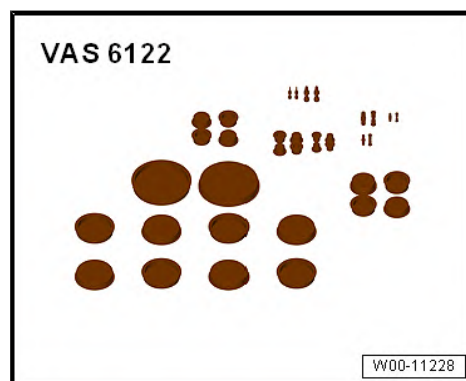
- ◆ Torque wrench - V.A.G 1331-



- ◆ Drip tray for workshop hoist - VAS 6208-



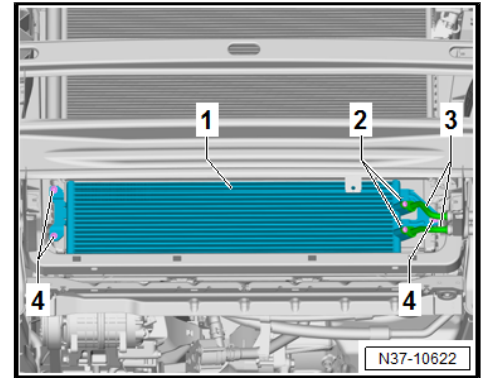
- ◆ Engine bung set - VAS 6122-





Removing

- Remove front bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Front bumper; Removing and installing bumper cover .
- Place drip tray for workshop hoist - VAS 6208- under ATF cooler.
- Unscrew bolts -2-
- Remove oil pressure lines -3-.
- Seal off open oil pressure lines and connections with clean plugs from engine bung set - VAS 6122- .
- Unscrew bolts -4-
- Remove ATF cooler -1-.



Installing

Install in reverse order of removal, observing the following:

- Renew oil seals in oil pressure lines ⇒ [Item 2 \(page 13\)](#) .
- Check ATF level ⇒ [page 20](#) .

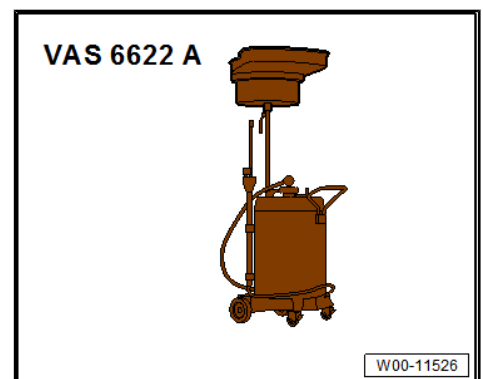
Specified torques

- ◆ ⇒ [“1.1 Assembly overview - ATF circuit”, page 13](#)

1.3 Removing and installing ATF filter

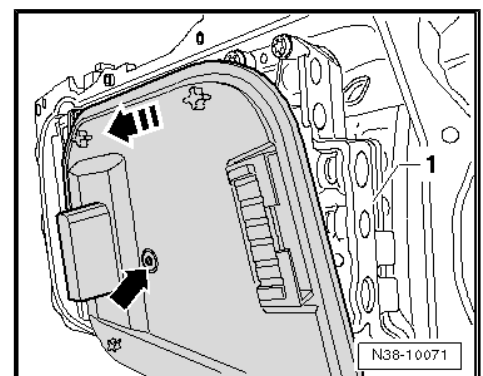
Special tools and workshop equipment required

- ◆ Used oil collector and extractor - VAS 6622A-



Removing

- Place used oil collection and extraction unit - VAS 6622A- below gearbox.
- Remove ATF sump ⇒ [page 52](#) .
- Pull off ATF filter -arrow- from gearbox -1- in direction of -arrow-.

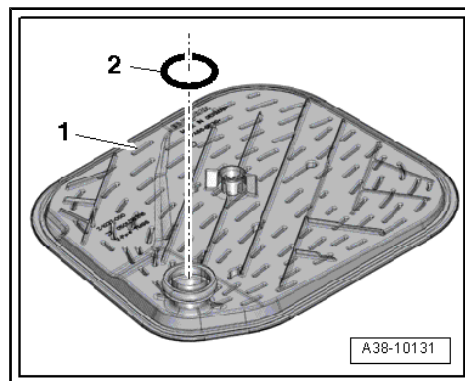




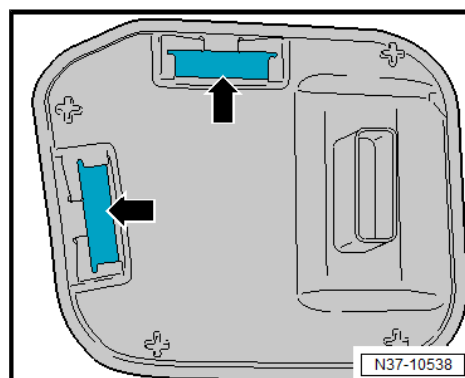
Installing

Install in reverse order of removal, observing the following:

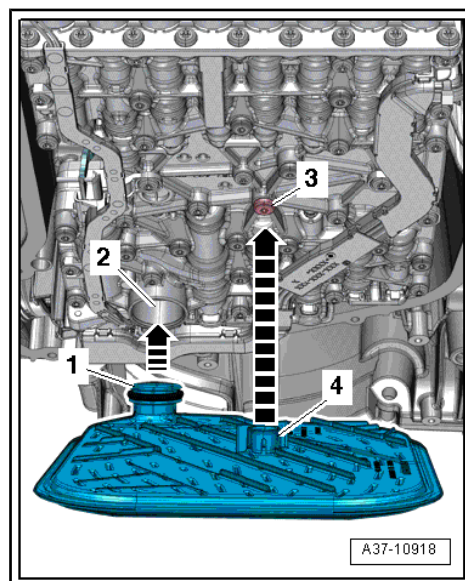
- Fit new O-ring -2- on ATF filter -1-.



- If fitted, clean magnets -arrows-.
- The solenoids must be firmly seated in guide.



- Fit ATF filter to mechatronic unit.
- ◆ Collar -1- of ATF filter must be inserted into opening -2- of mechatronic unit as far as stop.
- ◆ Receiver -4- must engage in bolt -3-.
- Install ATF sump ➔ [page 52](#) .
- Fill ATF ➔ [page 22](#) .

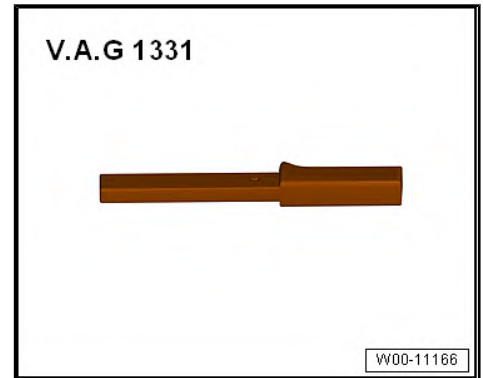


1.4 Removing and installing ATF oil pressure lines

Special tools and workshop equipment required



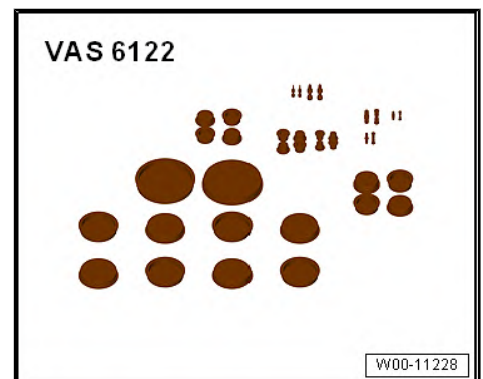
- ◆ Torque wrench - V.A.G 1331-



- ◆ Drip tray for workshop hoist - VAS 6208-



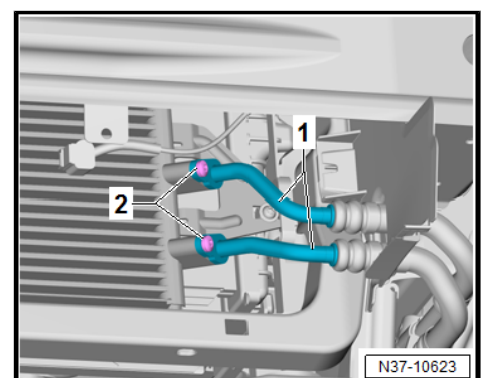
- ◆ Engine bung set - VAS 6122-



ATF oil pressure lines/ATF cooler to oil temperature regulator

Removing

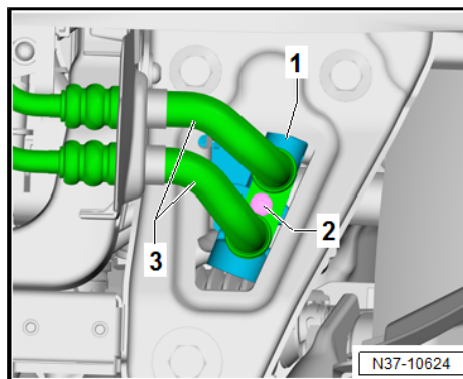
- Remove front bumper cover ⇒ General body repairs, exterior; Rep. gr. 63 ; Front bumper; Removing and installing bumper cover .
- Place drip tray for workshop hoist - VAS 6208- under ATF cooler.
- Unscrew bolts -2-
- Collect drained oil with drip tray for workshop hoists - VAS 6208- .
- Remove oil pressure lines -1-.
- Seal off open oil pressure lines and connections with clean plugs from engine bung set - VAS 6122- .





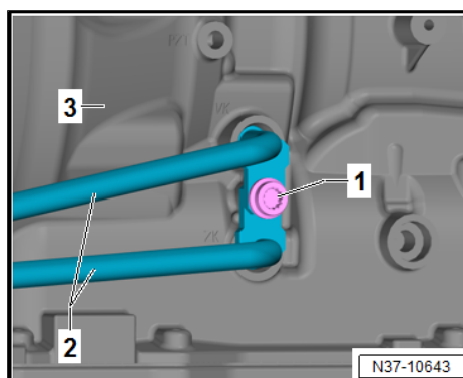
- Unscrew bolt -2-.
- Detach oil pressure lines -3- from oil temperature regulator -1-.
- Seal off open oil pressure lines and connections with clean plugs from engine bung set - VAS 6122- .

ATF oil pressure lines from gearbox to oil temperature regulator



Removing

- Place drip tray for workshop hoist - VAS 6208- under gearbox -3-.
- Unscrew bolt -3-.
- Detach oil pressure lines -2- from gearbox -3-.
- Seal off open oil pressure lines and connections with clean plugs from engine bung set - VAS 6122- .

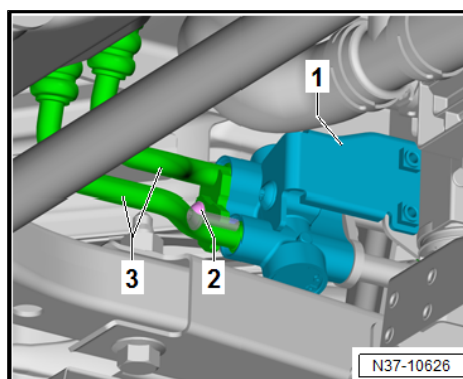


- Unscrew bolt -2-.
- Detach oil pressure lines -3- from oil temperature regulator -1-.
- Seal off open oil pressure lines and connections with clean plugs from engine bung set - VAS 6122- .

Installing

Install in reverse order of removal, observing the following:

- Renew oil seals in oil pressure lines ➔ [Item 2 \(page 13\)](#) .
- Check ATF level ➔ [page 20](#) .



Specified torques

➔ [“1.1 Assembly overview - ATF circuit”, page 13](#)

1.4.1 Removing and installing oil temperature regulator

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-

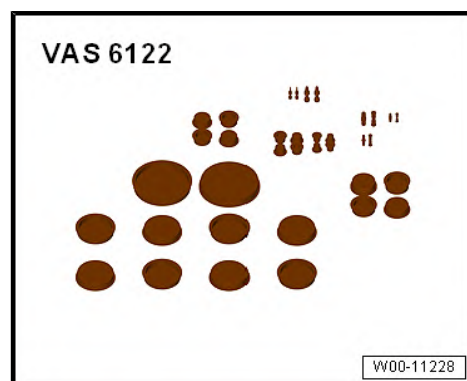




- ◆ Drip tray for workshop hoist - VAS 6208-



- ◆ Engine bung set - VAS 6122-



Removing

- Place drip tray for workshop hoist - VAS 6208- in area of oil temperature regulator -3-.
- Unscrew bolts -1-
- Detach lines -2-.
- Seal off open oil pressure lines and connections with clean plugs from engine bung set - VAS 6122- .
- Separate retainer for oil temperature regulator by unscrewing bolt ⇒ [Item 9 \(page 13\)](#) .

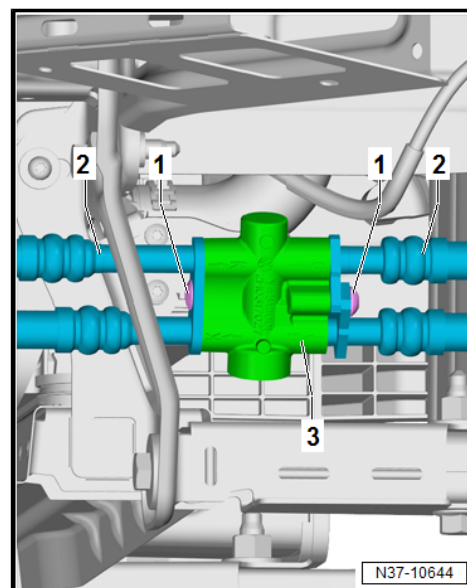
Installing

Install in reverse order of removal, observing the following:

- Renew oil seals in oil pressure lines ⇒ [Item 2 \(page 13\)](#) .
- Check ATF level ⇒ [page 20](#) .

Specified torques

⇒ [“1.1 Assembly overview - ATF circuit”, page 13](#)





2 ATF

⇒ [“2.1 Checking ATF level”, page 20](#)

⇒ [“2.2 Draining and filling ATF”, page 22](#)

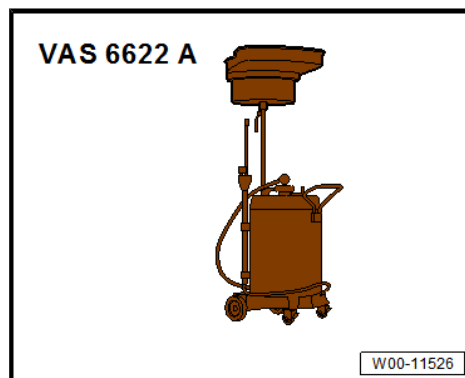
⇒ [“2.3 Purging ATF circuit”, page 26](#)

⇒ [“2.3.1 Bleeding cooling system of gearbox and leakage test”, page 28](#)

2.1 Checking ATF level

Special tools and workshop equipment required

- ◆ Used oil collector and extractor - VAS 6622A-



- ◆ Vehicle diagnostic tester



Caution

Make sure the vehicle is in a level position when checking, topping up or filling ATF. An improper ATF level hinders the function of the gearbox.

Prerequisites for testing

- Gearbox must not be in emergency running mode.
- Vehicle must be standing absolutely level.
- Gearbox is set to position “P”.
- Extraction hose/extraction hoses of a switched-on exhaust gas extraction system is/are connected.
- Heating and air conditioning system are switched off.
- ⇒ Vehicle diagnostic tester is connected.
- At the beginning of the test, the maximum ATF temperature should be 40 °C. If this temperature is exceeded, the gearbox first must cool off.



Note

- ◆ *Use ⇒ Vehicle diagnostic tester to read the ATF temperature.*
- ◆ *The further procedure is described in the ⇒ Vehicle diagnostic tester.*
- ◆ *The ATF level is affected by the ATF temperature.*
- ◆ *Checking the ATF level when the ATF temperature is too low leads to overfilling.*
- ◆ *Checking the ATF level when the ATF temperature is too high leads to underfilling.*
- ◆ *Both an excessive and an insufficient ATF level affects the function of the gearbox.*
- ◆ *The ATF level can be checked at the ATF inspection and filler hole.*
- ◆ *The ATF level is correct if ATF is slightly dripping out of ATF inspection/filler hole at an ATF temperature of 40°C (due to the increased ATF level at higher temperatures).*
- ◆ *Renew plug for ATF inspection and filler hole.*



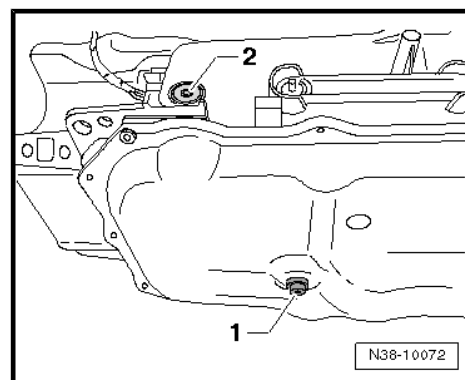
Caution

Risk of damage to gearbox.

- ◆ *Only the ATF ordered through the replacement part system can be used in automatic gearbox 0DR.*
- ◆ *Other oils will lead to malfunctions or failure of the gearbox.*
- ◆ *For allocation, see ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE*
- ◆ *The adapter for filling ATF must be clean. The ATF must not be mixed with other ATF oils!*

Observe test prerequisites:

- Parking brake applied
- Test temperature: max. 40°C
- Place used oil collection and extraction unit - VAS 6622A- below gearbox.
- At idling speed, unscrew ATF inspection plug -2-.





- Always renew inspection plug with seal -arrow-.

If ATF drips out of inspection hole:

ATF need not be topped up.

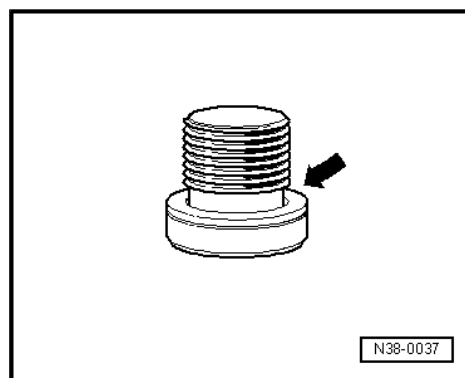
- Tighten new inspection plug to specified torque.

If no ATF drips out of inspection hole:

- ATF must be topped up ➔ [page 25](#) .

Torque settings

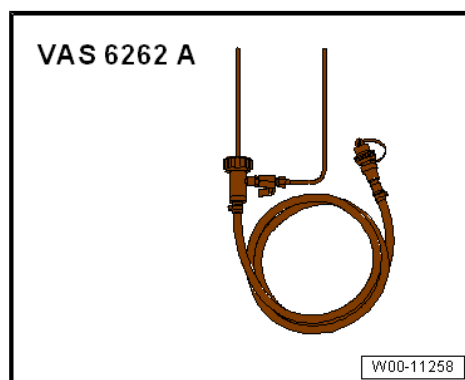
Component	Torque setting
ATF inspection plug	35 Nm



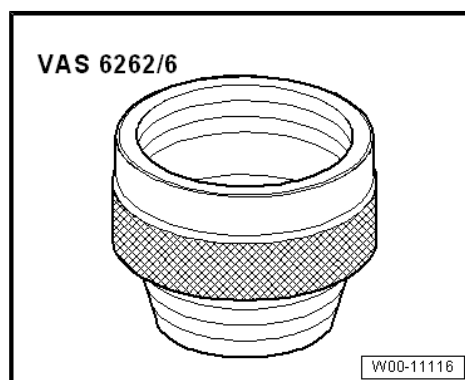
2.2 Draining and filling ATF

Special tools and workshop equipment required

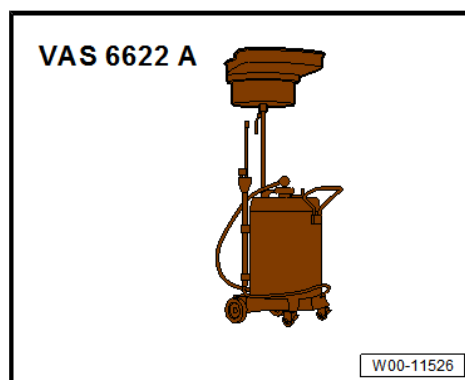
- ◆ Adapter for oil filling - VAS 6262 A-



- ◆ Adapter swivel hose - VAS 6262/5-
- ◆ Adapter - VAS 6262/6- , if necessary



- ◆ Used oil collector and extractor - VAS 6622A-



- ◆ Vehicle diagnostic tester



Do not mix any additives into the ATF oil or use a substitute oil.
Drained ATF oil must not be reused.



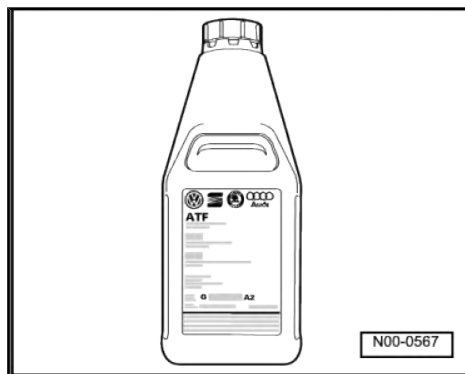
Fill only ATF available as a replacement part from ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis for MAN TGE.

Capacity ⇒ [page 6](#) .



Caution

Make sure the vehicle is in a level position when checking, topping up or filling ATF. An improper ATF level hinders the function of the gearbox.



Note

If the oil temperature is above 45°C, allow the gearbox to cool down.

Conditions

- Engine is not running.
- Gearbox must not be in emergency running mode.
- Vehicle must be standing absolutely level.
- Gearbox is set to position "P".
- Extraction hose/extraction hoses of a switched-on exhaust gas extraction system is/are connected.
- Heating and air conditioning system are switched off.
- ⇒ Vehicle diagnostic tester is connected.
- The ATF temperature should be max. 40°C at the start of work. If this temperature is exceeded, the gearbox first must cool off.



Note

- ◆ *Use ⇒ Vehicle diagnostic tester to read the ATF temperature.*
- ◆ *The further procedure is described in the ⇒ Vehicle diagnostic tester.*
- ◆ *The ATF level is affected by the ATF temperature.*
- ◆ *Checking the ATF level when the ATF temperature is too low leads to overfilling.*
- ◆ *Checking the ATF level when the ATF temperature is too high leads to underfilling.*
- ◆ *Both an excessive and an insufficient ATF level affects the function of the gearbox.*
- ◆ *The ATF level can be checked at the ATF inspection and filler hole.*
- ◆ *The ATF level is correct if ATF is slightly dripping out of ATF inspection/filler hole at an ATF temperature of 40°C (due to the increased ATF level at higher temperatures).*
- ◆ *Renew plug for ATF inspection/filler hole.*



Caution

Do not start engine if there is no oil in gearbox.



WARNING

Risk of eye injury.

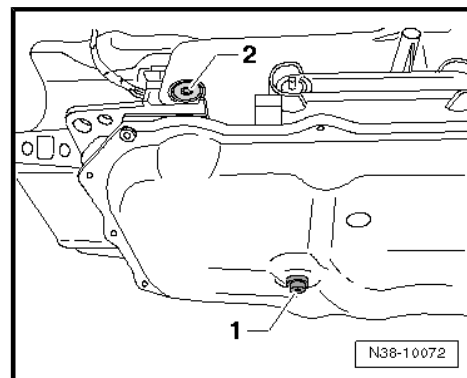
- ◆ *Wear safety goggles.*

Draining ATF

- Place used oil collection and extraction unit - VAS 6622A- below gearbox.
- Remove ATF drain plug -1- from sump.
- Drain ATF.

Adding ATF

- Tighten ATF drain plug in sump.



Caution

Make sure the vehicle is in a level position when checking, topping up or filling ATF. An improper ATF level hinders the function of the gearbox.

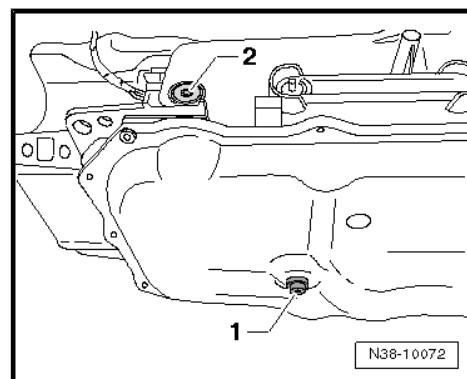


Caution

Risk of damage to gearbox.

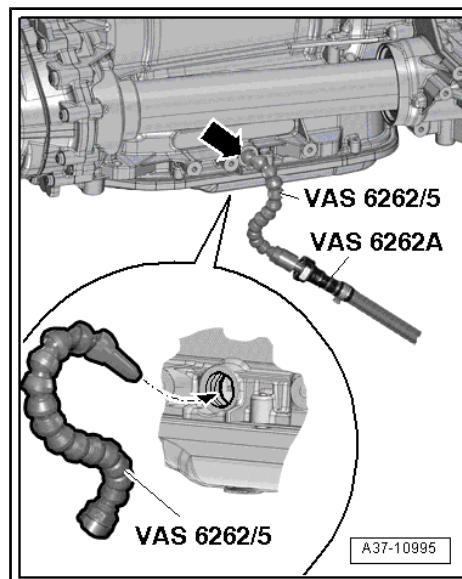
- ◆ *Only the ATF ordered through the replacement part system can be used in automatic gearbox 0DR.*
- ◆ *Other oils will lead to malfunctions or failure of the gearbox.*
- ◆ *For allocation, see ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE*
- ◆ *The adapter for filling ATF must be clean. The ATF must not be mixed with other ATF oils!*

- Raise and tie up oil container holder with adapter for oil filling - VAS 6262 A- as high up on the vehicle as possible.
- Unscrew inspection plug -2-.

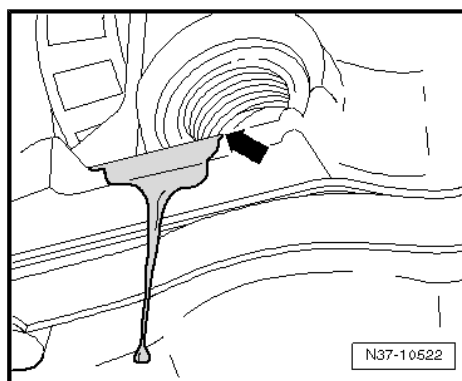




- Insert adapter for ATF oil filling - VAS 6262/5- into ATF inspection and filler hole -arrow-.
- Pour in ATF through adapter until ATF runs out of inspection and filler hole.



- Start engine and immediately fill more ATF until it flows out of overflow edge -arrow-.
- Screw in plug for ATF inspection/filler hole hand-tight.
- Run engine at idling speed.
- With brake pressed, shift through all selector lever positions at idling speed and make sure each position is engaged for at least 10 seconds.
- Switch selector lever to “P” position.
- Keep constant engine speed of 2000 rpm for 30 seconds to allow torque converter to be filled.
- Allow gearbox to cool down to test temperature (40°C), check ATF level and top up if necessary.
- Tighten ATF inspection plug.
- Carry out a road test to bring ATF temperature up to at least 80°C and to open ATF radiator thermostat.
- Allow gearbox to cool down to test temperature (40°C), check ATF level and top up if necessary.
- Tighten ATF inspection plug.



Torque settings

Component	Torque setting
ATF oil drain plug	12 Nm
ATF inspection plug	35 Nm

2.3 Purging ATF circuit

If ATF is dirty or if an exchange gearbox is to be installed, clean ATF cooler and ATF lines with compressed air (maximum 10 bar).

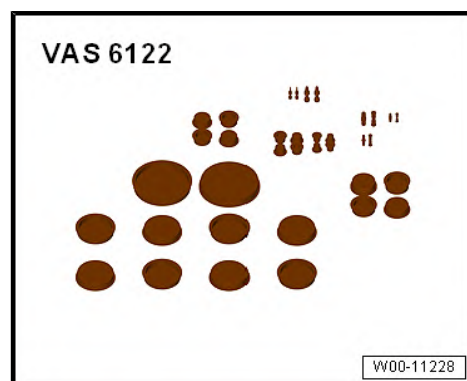
Special tools and workshop equipment required



- ◆ Drip tray for workshop hoist - VAS 6208-



- ◆ Engine bung set - VAS 6122-



Removing



WARNING

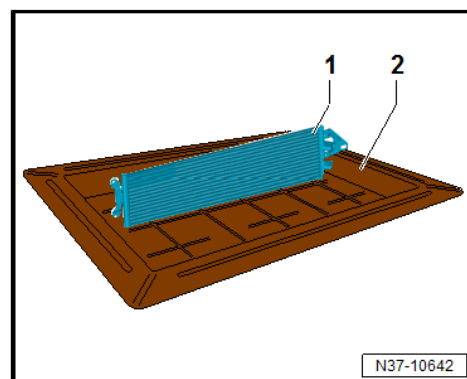
Wear safety goggles.



Note

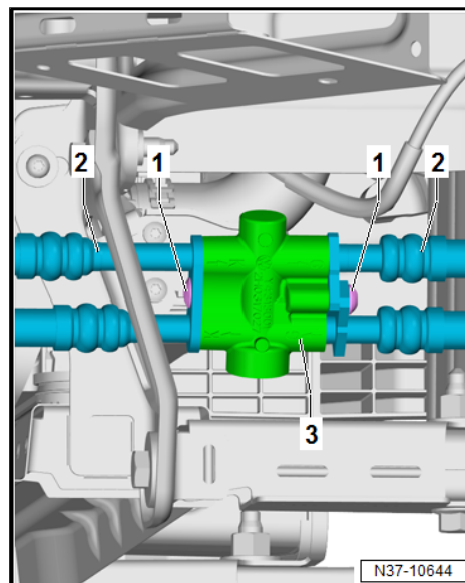
Dispose of used oil properly.

- Thoroughly clean area around unions for gearbox to ATF line, ATF line to oil temperature regulator and ATF line to ATF cooler.
- Remove ATF cooler → [page 14](#) .
- Empty ATF cooler -1-. Use drip tray for workshop hoist - VAS 6208- -2- to do this.

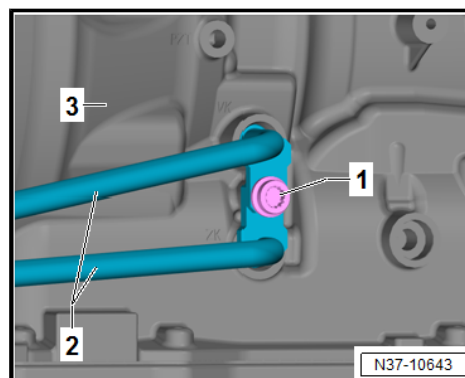




- Place drip tray for workshop hoist - VAS 6208- beneath oil temperature regulator -3-.
- Unscrew bolts -1-
- Pull oil pressure lines -2- out of oil temperature regulator -3-.



- Unscrew bolt -1-.
- Detach oil pressure lines -2- from gearbox -3-.
- Seal holes in gearbox with clean plugs from engine bung set - VAS 6122- .
- Fit hose onto oil pressure line of gearbox/oil temperature regulator and secure with a hose clip.
- Insert end of hose in a suitable container.
- Clean oil pressure line compressed air.
- Push hose onto other oil pressure line and repeat procedure.



Installing

Install in reverse order of removal, observing the following:

- Renew seals ⇒ [Item 4 \(page 13\)](#) .

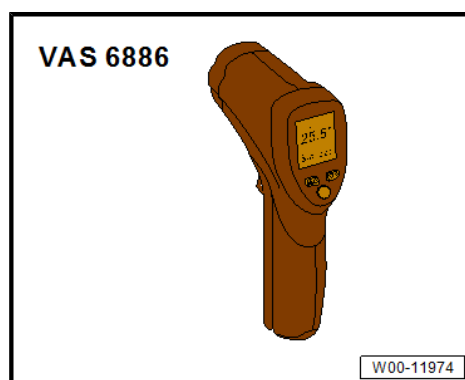
Specified torques

- ♦ ⇒ [“1.1 Assembly overview - ATF circuit”, page 13](#)

2.3.1 Bleeding cooling system of gearbox and leakage test

Special tools and workshop equipment required

- ♦ Temperature gauge - VAS 6886-



Sequence of operations

- Start engine and allow to idle.



- Check temperature of gear oil cooler with temperature gauge
- VAS 6886- .
- Oil temperature regulator opens at 70°C



Note

- ◆ *Bleed system during this time.*
- ◆ *Allow warm engine to idle for ten minutes.*
- ◆ *Check for leaks in lines through which media flow.*

Concluding measures

- Switch off engine, and let it cool off.
- Check ATF level ⇒ [page 20](#) .



3 Removing and installing gearbox

⇒ ["3.1 Specified torques for gearbox", page 30](#)

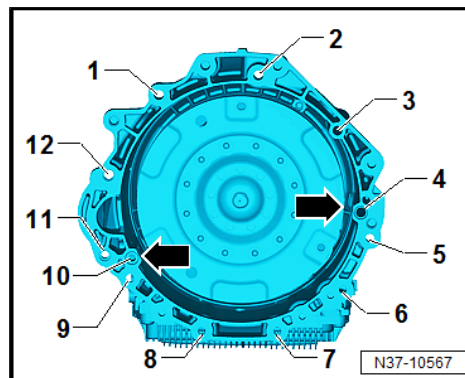
⇒ ["3.2 Removing gearbox", page 30](#)

⇒ ["3.3 Installing gearbox", page 34](#)

3.1 Specified torques for gearbox

Specified torques for gearbox

Component	Torque setting
-Item 1- and -item 2- M12 × 70	60 Nm
-Item 3- and -item 4- M12 × 230	60 Nm
-Item 5-, -item 6-, -item 7-, -item 8-, -item 9- M10 × 65	45 Nm
-Item 10- M12 × 70	60 Nm
-Item 11- M 12 × 60 - M 8 × 12 (centre hex studs, starter)	60 Nm
Item -12- M 12 × 70 (starter bolts)	60 Nm
-Arrows-	Dowel sleeves for centring engine/ gearbox in cylinder block



3.2 Removing gearbox

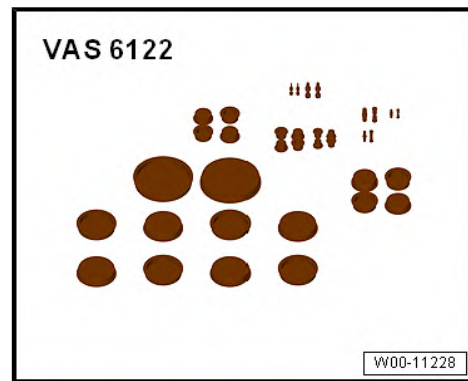
Special tools and workshop equipment required

- ♦ Engine and gearbox jack - VAS 6931-





- ◆ Engine bung set - VAS 6122-



- ◆ Tensioning belt - T10038- , qty. 2



- ◆ Support bridge - 30-211A-



- ◆ Drip tray for workshop hoist - VAS 6208-

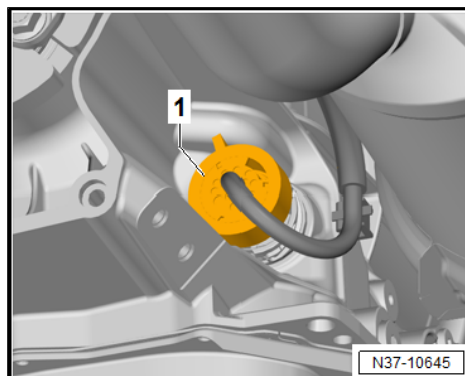


Sequence of operations

- Move selector lever to position P.
- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .

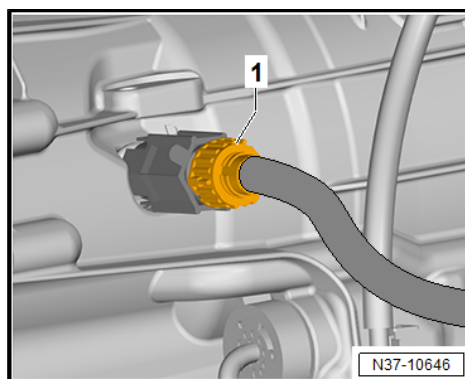


- Remove subframe together with steering rack ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Removing and installing subframe with steering rack .
- Remove starter ⇒ Electrical system; Rep. gr. 27 ; Starter; Removing and installing starter .
- Remove catalytic converter ⇒ Rep. gr. 26 ; Emission control; Removing and installing catalytic converter .
- Remove front propshaft ⇒ Propshaft and rear final drive; Rep. gr. 39 ; Propshaft; Removing and installing propshaft .
- Disconnect electrical connector -1-.



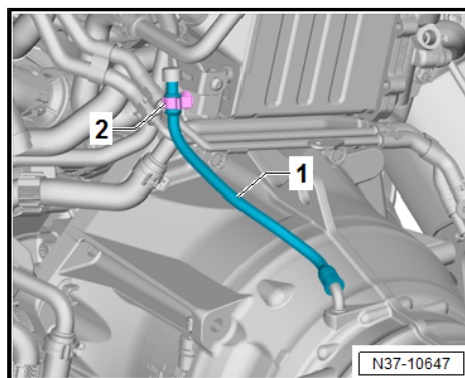
Only vehicles with tachograph

- Disconnect electrical connector -1-.



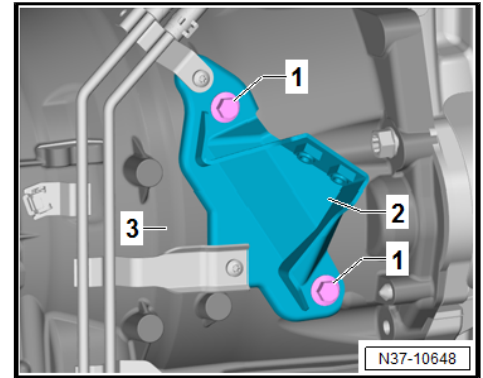
All vehicles

- Remove hose -1- from clip -2-.

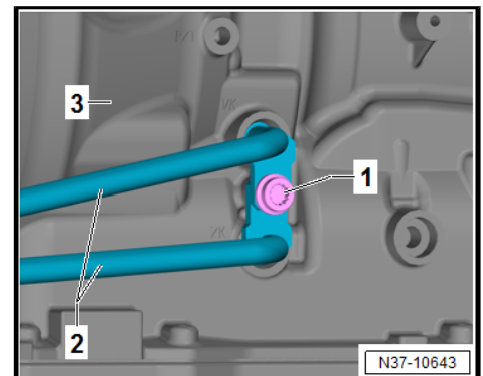




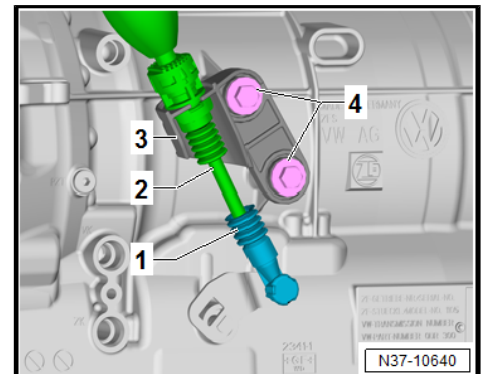
- Unscrew bolts -1-
- Remove bracket -2- from gearbox -3-.



- Unscrew bolt -1-.
- Collect drained oil with drip tray for workshop hoists - VAS 6208- .
- Detach oil pressure lines -2- from gearbox -3-.
- Seal holes in gearbox and oil pressure lines with clean plugs from engine bung set - VAS 6122- .



- Detach ball head -1- from selector lever.
- Unscrew bolts -4-
- Place cable support bracket -3- with selector lever cable -2- to one side together with servo oil reservoir and secure in place.

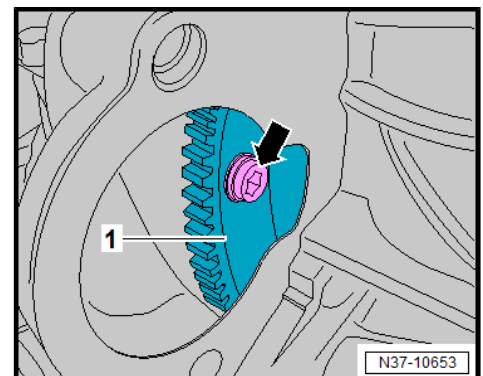


- Remove starter ⇒ Electrical system; Rep. gr. 27 ; Starter; Removing and installing starter .



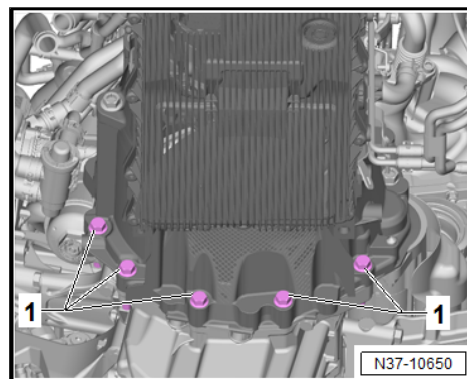
Note

- ◆ *Only turn crankshaft in direction of engine rotation.*
- ◆ *At every 120° crankshaft revolution there is a bolted connection.*
- Turn crankshaft by vibration damper until bolt -arrow- in drive plate -1- is accessible.
- Unscrew bolt -arrow-.

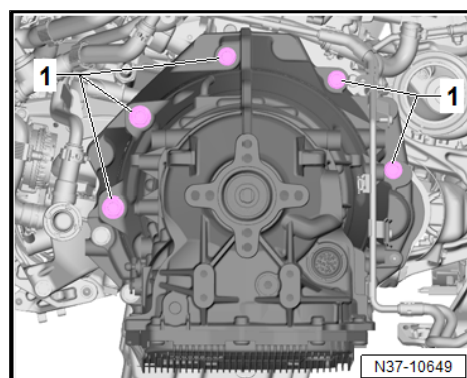




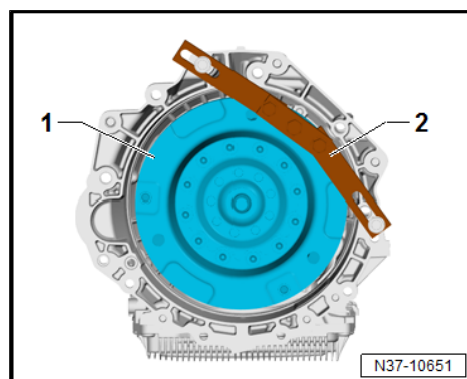
- Unscrew bolts -1-
- Support gearbox with foam underlay on engine and gearbox jack - VAS 6931- .
- Secure gearbox with two tensioning straps - T10038- .
- Lower side of gearbox slightly.



- Unscrew bolts -1-
- A second mechanic is required to assist removal of the gearbox.
- Press gearbox off engine (dowel sleeves) and pull out to rear.
- Lower engine and gearbox jack - VAS 6931- while a second mechanic guides the gearbox.



- Secure torque converter -1- with support bridge - 30-211A-2-.



3.3 Installing gearbox

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





Note

For information about which other tools are needed, see Removing gearbox ➔ [page 30](#) .

Installing

Install in reverse order of removal, observing the following:



Note

All dowel sleeves for aligning engine and gearbox must be present.



Caution

When doing any repair work, especially in the engine compartment, pay attention to the following due to the cramped conditions:

- ◆ *Route lines of all kinds and electrical cables so as to re-establish the original wiring.*
- ◆ *Ensure sufficient clearance to all moving or hot components.*



- If present, remove support bridge - 30-211A- -1- from torque converter -2-.



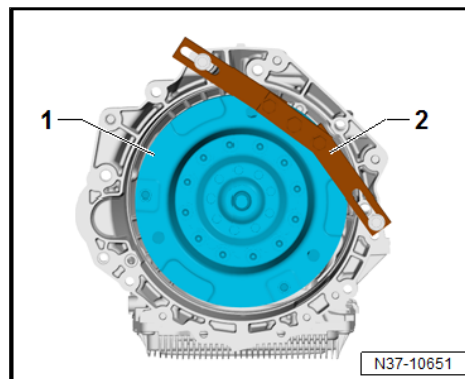
Caution

If engine and gearbox are brought together with force because the torque converter is improperly fitted, both gearbox and torque converter will be damaged.

- Check torque converter is properly seated.
- The torque converter is properly installed if it can be turned easily by hand and sits around the whole circumference at an equal depth in the gearbox.
- Align connecting points to torque converter and follower before joining.
- Turn crankshaft to check that torque converter is not jammed.
- Connecting points aligned to torque converter and follower.
- Tighten bolts that secure torque converter to follower by hand.
- Join gearbox to engine.
- Adjust selector lever cable ➔ [page 50](#) .
- Check ATF level ➔ [page 20](#) .

Specified torques

- ◆ Assembly mountings; Assembly overview - assembly mountings ➔ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings .
- ◆ Noise insulation; Removing and installing noise insulation ➔ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation
- ◆ Battery; Disconnecting and connecting battery ➔ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and connecting battery
- ◆ ➔ [“3.1 Specified torques for gearbox”, page 30](#)
- ◆ Starter ➔ Electrical system; Rep. gr. 27 ; Starter
- ◆ Exhaust system; Emission control; Assembly overview - emission control ➔ Rep. gr. 26 ; Exhaust system; Emission control; Assembly overview
- ◆ ➔ [“1.1 Assembly overview - torque converter”, page 9](#)
- ◆ Propshaft; Assembly overview - propshaft ➔ Propshaft and rear final drive; Rep. gr. 39 ; Propshaft; Assembly overview - propshaft
- ◆ ➔ [“7.1 Assembly overview - selector mechanism”, page 43](#)





4 Assembly mountings

⇒ [“4.1 Assembly overview - assembly mountings”, page 37](#)

⇒ [“4.2 Assembly overview - gearbox carrier with gearbox mounting”, page 37](#)

⇒ [“4.3 Removing and installing gearbox mounting”, page 38](#)

⇒ [“4.4 Removing and installing gearbox carrier”, page 38](#)

4.1 Assembly overview - assembly mountings

Assembly overview - assembly mountings for longitudinally installed engine and gearbox ⇒ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings .

4.2 Assembly overview - gearbox carrier with gearbox mounting

1 - Bolts

- ☐ Renew after removal
- ☐ Qty. 2

2 - Bolts

- ☐ Qty. 2
- ☐ Specified torque ⇒ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings

3 - Gearbox mounting

- ☐ Removing and installing ⇒ Rep. gr. 10 ; Assembly mountings; Removing and installing gearbox mounting

4 - Bolts

- ☐ Qty. 4
- ☐ Specified torque ⇒ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings

5 - Bevel box carrier

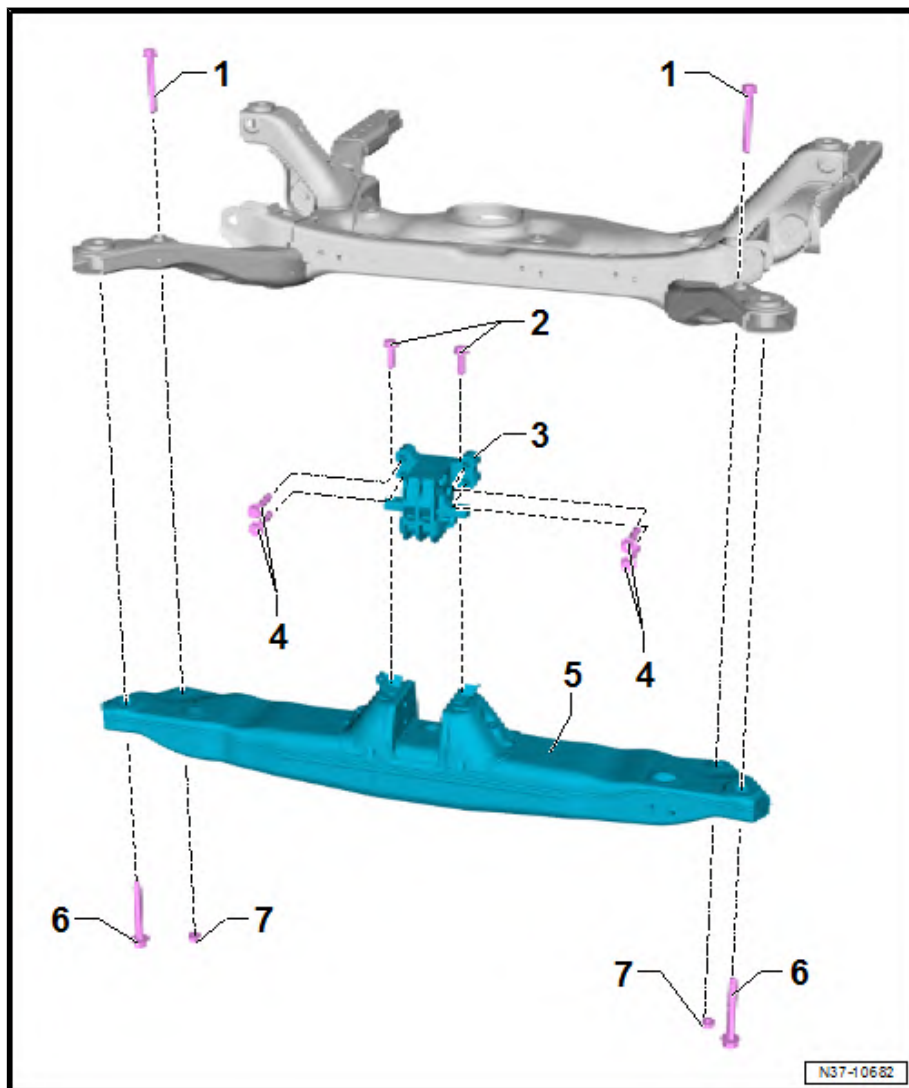
- ☐ Removing and installing ⇒ [page 38](#)

6 - Bolts

- ☐ Qty. 2
- ☐ Specified torque ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Assembly overview - subframe

7 - Nuts

- ☐ Renew after removal
- ☐ Qty. 2
- ☐ 50 Nm +180°





4.3 Removing and installing gearbox mounting

- Removing and installing gearbox mounting ⇒ Rep. gr. 10 ; Assembly mountings; Removing and installing gearbox mounting

4.4 Removing and installing gearbox carrier

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-

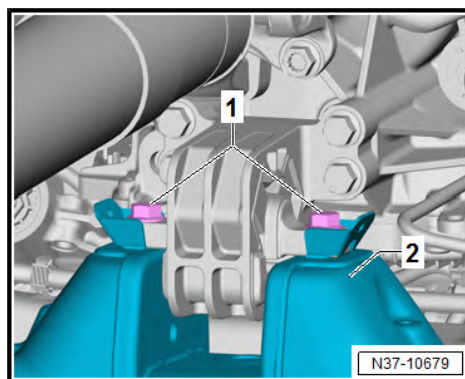


- ◆ Retaining tool - VW785/1B-

- ◆ -Support-

Removing

- Prop up gearbox with -support- or take up weight with retaining tool - VW785/1B- .
- Unscrew bolts -1-
- Using -support- or retaining tool - VW785/1B- , lightly lift off gearbox from carrier -2-.

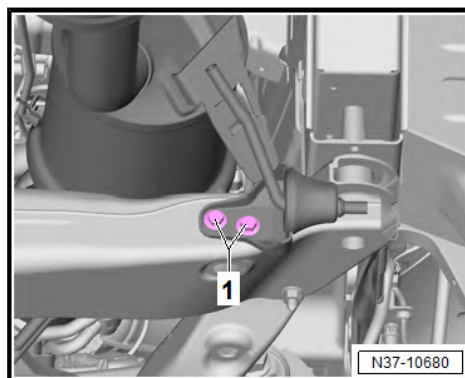


- Unscrew bolts -1-



Note

Assistance of a second mechanic may be necessary for this.





- Unscrew bolts -1-
- Unscrew nuts -2-. Remove carrier -3- when doing this.

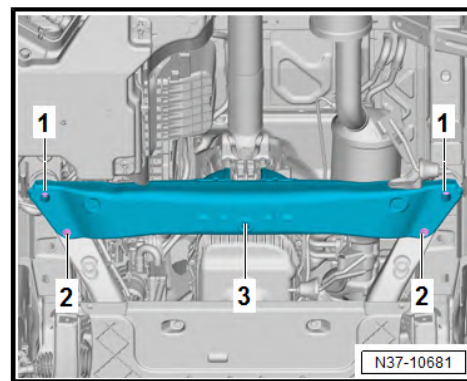
Installing

Install in reverse order of removal, observing the following:

- Align gearbox mountings free of stress before tightening to specified torque.

Specified torques

- ◆ ⇒ [“4.2 Assembly overview - gearbox carrier with gearbox mounting”, page 37](#)





5 Transporting gearbox

Special tools and workshop equipment required

- ◆ Workshop hoist - VAS 6100-



- ◆ Support bridge - 30-211A-



- ◆ Lifting tackle - 3033-



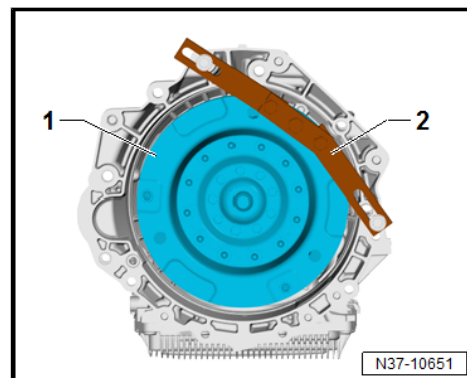
- ◆ Shackle - 10-222A/12-



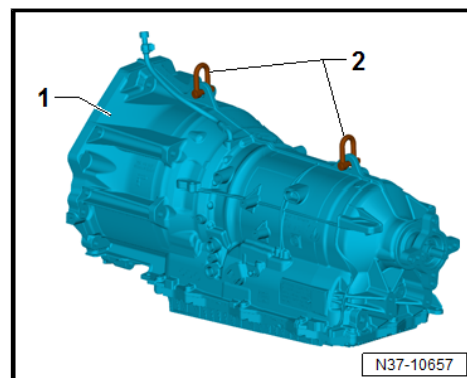


Sequence of operations

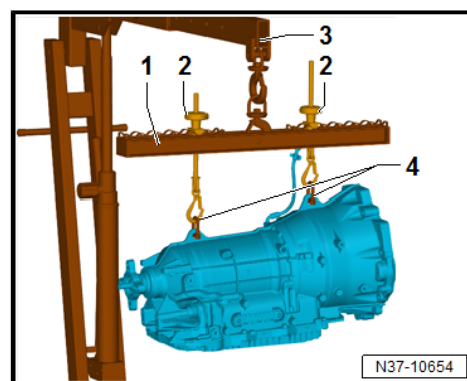
- Secure torque converter -1- with support bridge - 30-211A- -2-.



- Secure shackle - 10-222A/12- -2- on gearbox -1-.



- Attach lifting tackle - 3033- -1- to workshop hoist - VAS 6100- -3-.
- Hook lifting tackle - 3033- -1- into shackles -4- and take up weight slightly.
- If necessary, adjust position of gearbox via spindles -2-.





6 Securing to engine and gearbox support

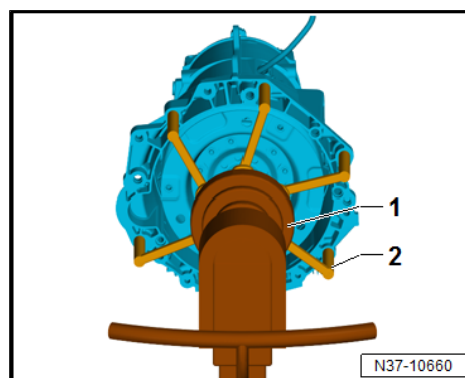
Special tools and workshop equipment required

- ◆ Engine and gearbox support - VAS 6095A-



Sequence of operations

- Remove gearbox ⇒ [page 30](#) .
- Transporting gearbox ⇒ [page 40](#) .
- If necessary, remove torque converter ⇒ [page 9](#) .
- Secure gearbox with mountings -2- on engine and gearbox support - VAS 6095A- -1-.





7 Selector mechanism

⇒ [“7.1 Assembly overview - selector mechanism”, page 43](#)

⇒ [“7.2 Checking selector mechanism”, page 44](#)

⇒ [“7.3 Emergency release from position P”, page 45](#)

⇒ [“7.4 Removing and installing selector mechanism”, page 45](#)

⇒ [“7.5 Removing and installing selector lever handle”, page 47](#)

⇒ [“7.6 Removing and installing gear selector cable”, page 48](#)

⇒ [“7.7 Checking and adjusting selector lever cable”, page 50](#)

7.1 Assembly overview - selector mechanism

1 - Selector lever handle

- ☐ With selector cover
- ☐ Removing and installing
⇒ [page 47](#)

2 - Clip

- ☐ Renew after removal

3 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

4 - Selector mechanism

- ☐ With lever for manual re-release mechanism
- ☐ Carrying out emergency release ⇒ [page 45](#) .
- ☐ Removing and installing
⇒ [page 45](#)

5 - Adjuster screw

- ☐ For adjusting selector lever cable
- ☐ 12 Nm

6 - Bolts

- ☐ For selector lever cable
- ☐ Qty. 2
- ☐ 12 Nm

7 - Tunnel seal

- ☐ On selector lever cable

8 - Nuts

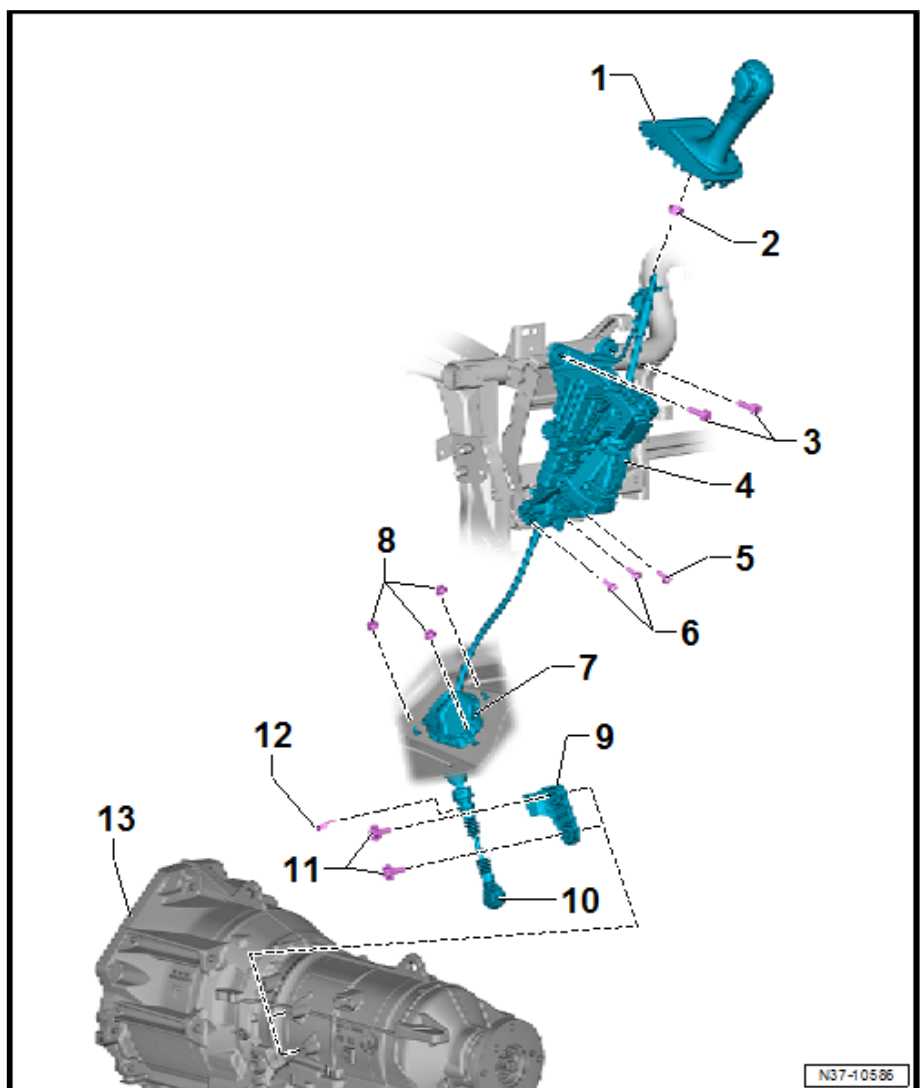
- ☐ Qty. 3
- ☐ 8 Nm

9 - Cable support bracket

- ☐ For selector lever cable to gearbox

10 - Selector lever cable

- ☐ Do not bend or kink
- ☐ Do not grease
- ☐ Removing and installing ⇒ [page 48](#)





- ☐ Adjusting ⇒ [page 50](#)

11 - Bolts

- ☐ Qty. 2
- ☐ 20 Nm

12 - Securing clip

- ☐ For selector lever cable to cable support bracket.

13 - Gearbox

- ☐ Removing ⇒ [page 30](#)
- ☐ Installing ⇒ [page 34](#)

7.2 Checking selector mechanism

- Move selector lever to P position.
- Switch on ignition.
- Do not press brake pedal.
- Selector lever is locked and cannot be shifted out of P position with lock button on selector lever handle pressed. Solenoid for selector lever lock - N110- blocks selector lever.
- Depress brake pedal and hold.
- Solenoid for selector lever lock - N110- releases selector lever. It is possible to shift into a driving gear. With lock button on selector lever handle pressed, slowly shift selector lever through positions R, N, D, S starting with position P. Check whether selector lever position display - Y6- in dash panel insert corresponds to selector lever position.
- Move selector lever to P position.
- Switch on ignition.
- Do not press brake pedal.
- After a brief wait: selector lever is locked and cannot be shifted out of N position despite lock button on selector lever handle being pressed. Solenoid for selector lever lock - N110- blocks selector lever.
- Press brake pedal.
- Solenoid for selector lever lock - N110- releases selector lever. D position can be engaged.
- Move selector lever to D position.
- Switch on ignition.
- Shift selector lever into Tiptronic gate.
- Illuminated D symbol in selector lever position display - Y6- must go out and + and – symbols must light up.

Selector lever position indicator

Simultaneous illumination of all selector lever position display segments indicates gearbox emergency running mode.

- Check and adjust selector lever cable ⇒ [page 50](#) .



7.3 Emergency release from position P

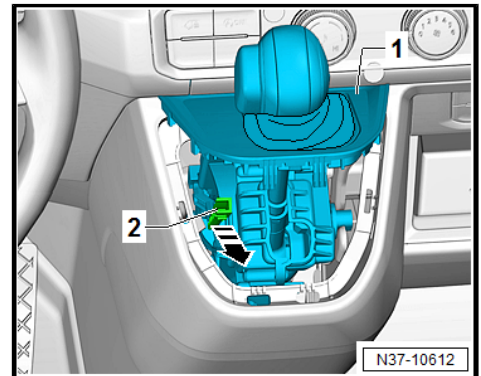
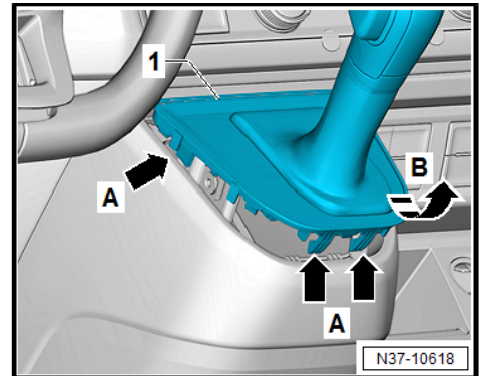
Procedure



Note

Operating emergency release causes selector lever lock solenoid - N110- to release lock when it is de-energised.

- Unclip selector cover -1- on »locking lugs« -arrows A- and pull out in -direction of arrow B-.
- Only raise selector cover -1- far enough for manual release mechanism lever to become accessible.
- Press release mechanism lever -2- in direction of -arrow-. At the same time, press button on handle and move gear selector lever out of P position.
- Install selector cover.



7.4 Removing and installing selector mechanism

Special tools and workshop equipment required

- ◆ Torque wrench - V.A.G 1331-





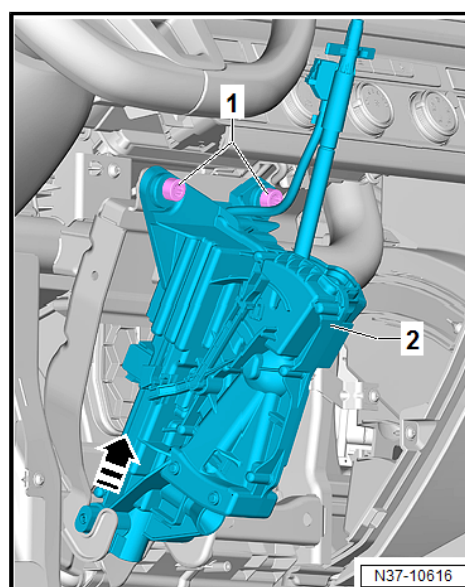
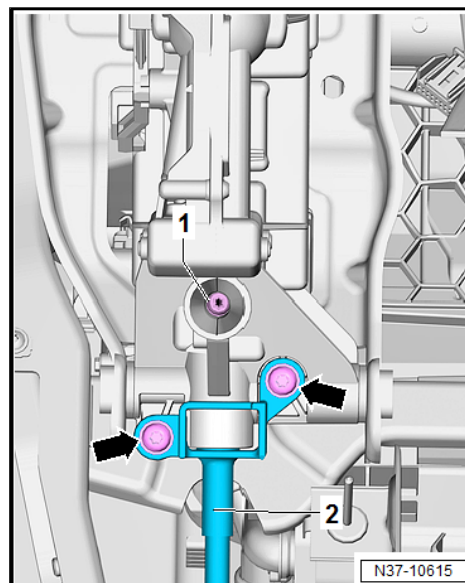
Removing

- Switch selector lever to P position.
- Remove selector lever handle ⇒ [page 47](#) .
- Disconnect battery ⇒ Electrical system; Rep. gr. 27 ; Battery; Disconnecting and reconnecting battery .



Note

- ◆ *Do not completely remove the dash panel.*
- ◆ *Detach dash panel enough so that selector mechanism can be guided passed it.*
- Detach dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .
- Loosen bolt for selector lever cable -1-.
- Unscrew bolts for cable support bracket -arrow-.
- Pull selector lever cable -2- with cable support bracket downwards out of selector mechanism.
- Unscrew bolts for selector mechanism -1-.
- Remove selector mechanism -2- upwards in direction of -arrow- from retainers on subframe.



- Release and disconnect electrical connector -2- on rear of selector mechanism -1-.

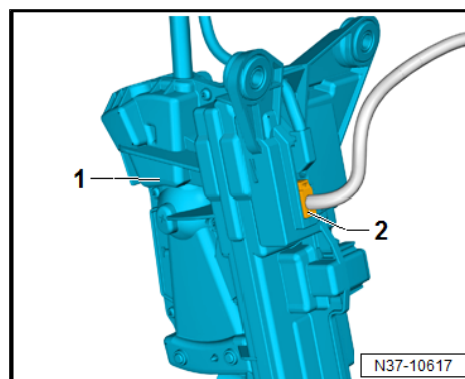
Installing

Install in reverse order of removal, observing the following:

- Check and adjust selector lever cable ⇒ [page 50](#) .
- Secure dash panel ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Removing and installing dash panel .
- Install selector lever handle ⇒ [page 47](#) .

Torque settings

- ◆ ⇒ [“7.1 Assembly overview - selector mechanism”, page 43](#)
- ◆ ⇒ General body repairs, interior; Rep. gr. 70 ; Dash panel; Assembly overview - dash panel





7.5 Removing and installing selector lever handle

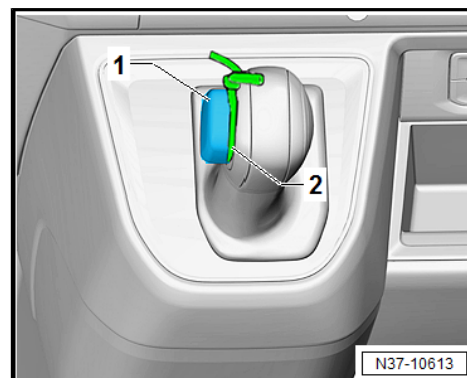
Special tools and workshop equipment required

- ◆ Hose clip pliers - V.A.G 1275 A-



Removing

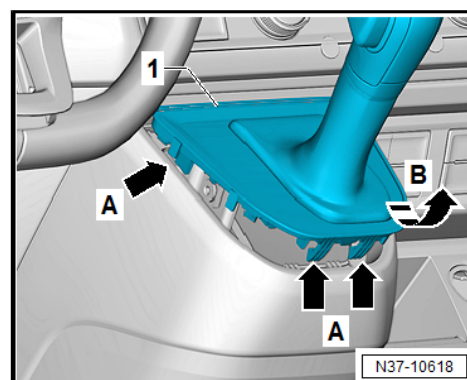
- Shift selector lever to D position.
- Switch off ignition.
- Apply handbrake.
- Carefully pull button -1- out of handle until a slight resistance can be felt and button engages in handle.
- Secure button -1- as shown.



Note

If there is no transport guard, the button can be secured with a thin cable tie or wire -2-.

- Unclip selector cover -1- on locking lugs -arrows A- and pull out in direction of -arrow B-.





- Release and disconnect electrical connector -1- from selector lever position display - Y6- .
- Remove clamp -2-.
- Pull the handle off the selector lever together with the boot and the cover. Do not press button when doing this.



Note

Handle is removed together with selector cover.

Installing

Install in reverse order of removal, observing the following:

- Selector lever is in D position.



Note

- ♦ *The button might have been pressed into the handle. Do not install a handle with the button pressed in.*
- ♦ *New handle is supplied with installation guard. Only remove securing clip immediately prior to installation.*

If the button was pressed in by mistake, the installation position can be restored. To do this, push installation guard fully in whilst button is pressed, release button and pull installation guard out again. Button engages in installation position.

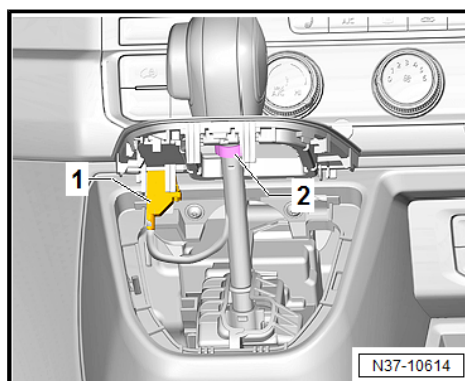
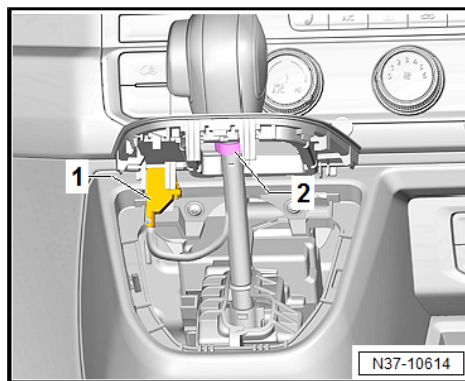
- Push handle with new clamp fully onto selector lever until it can be felt to engage.
- Press button in selector lever handle.



Note

If not installed correctly, button remains inserted in handle after being pressed. Handle must be re-installed.

- Attach new clamp -2- using hose clip pliers - V.A.G 1275- .
- Connect electrical connector -1- to selector lever position display - Y6- .



7.6 Removing and installing gear selector cable

Special tools and workshop equipment required



- ◆ Torque wrench - V.A.G 1331/-

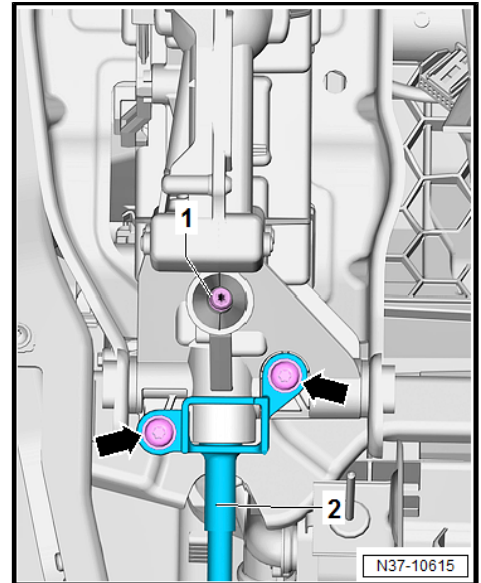
V.A.G 1331



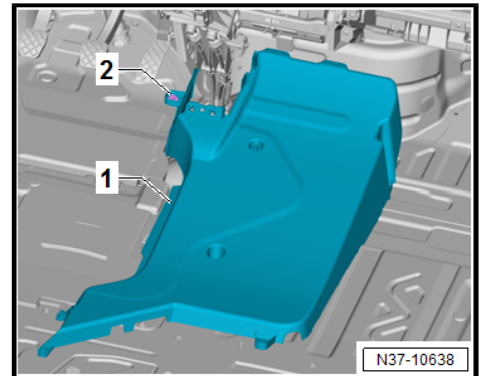
W00-11166

Removing

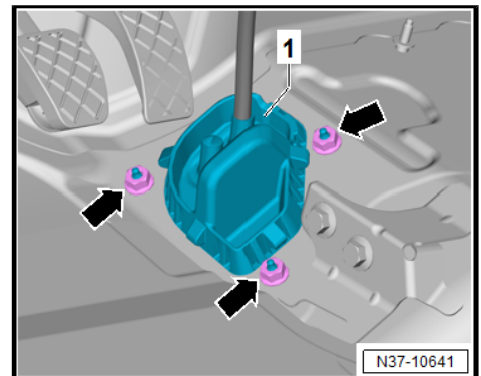
- Remove centre console ⇒ General body repairs, interior; Rep. gr. 68 ; Centre console; Removing and installing centre console .
- Remove centre console trim in footwell at front ⇒ General body repairs, interior; Rep. gr. 68 ; Centre console; Removing and installing centre console trim in front footwell .
- Remove floor covering on driver side ⇒ General body repairs, interior; Rep. gr. 70 ; Trims, interior; Removing and installing floor covering .
- Unscrew bolt -1-.
- Remove bolts -arrows-.
- Remove selector lever cable -2-.



- Unscrew nut -2-.
- Push footwell cover -1- aside in area of nut -2-.



- Remove nuts -arrows-.
- Detach tunnel seal -1-.





- Detach ball head -1- from selector lever.
- Unscrew bolts -4-
- Remove cable support bracket -3- with selector lever cable -2- downwards.

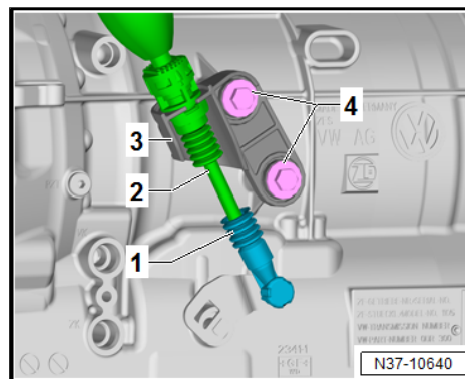
Installing

Install in reverse order of removal, observing the following:

- Check and adjust selector lever cable ⇒ [page 50](#) .

Specified torques

- ♦ ⇒ [“7.1 Assembly overview - selector mechanism”, page 43](#)



7.7 Checking and adjusting selector lever cable

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-



Sequence of operations

- Move selector lever to P position.
- Remove centre console ⇒ General body repairs, interior; Rep. gr. 68 ; Centre console; Removing and installing centre console .
- Loosen bolt -1- until selector lever cable -2- is free to move.
- Detent lever in gearbox must be engaged in parking lock.



Note

Block flange shafts of wheels in parking lock position.

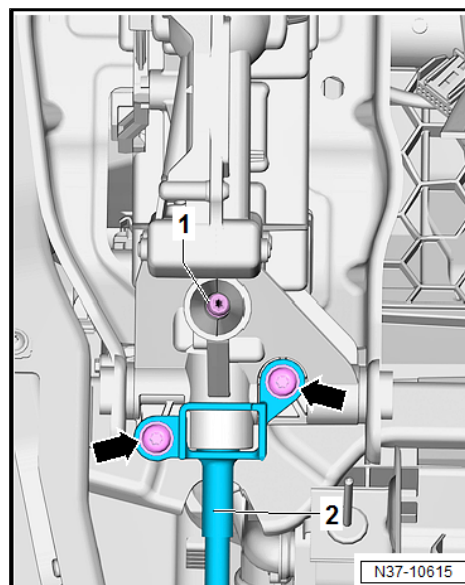
- Tighten bolt -1-.

Assembling

Assembly is carried out in reverse sequence; note the following:

Specified torques

- ♦ ⇒ [“7.1 Assembly overview - selector mechanism”, page 43](#)





38 – Gears, control

1 ATF system

⇒ [“1.1 Assembly overview - sump/oil strainer”, page 51](#)

⇒ [“1.2 Removing and installing sump”, page 52](#)

⇒ [“1.3 Removing and installing hydraulic impulse oil storage with accumulator solenoid N485”, page 54](#)

1.1 Assembly overview - sump/oil strainer

1 - Valve body

- ☐ Removing and installing
⇒ [page 60](#)

2 - Bolt

- ☐ Renew after removal
- ☐ Not available separately
- ☐ Observe tightening sequence
- ☐ 8 Nm

3 - Seal

- ☐ Renew after removal
- ☐ Not available separately

4 - Hydraulic impulse accumulator with accumulator solenoid - N485-

- ☐ Removing and installing
⇒ [page 54](#)

5 - Bolt

- ☐ Renew after removal
- ☐ Not available separately
- ☐ 8 Nm

6 - Seal

- ☐ Renew after removal
- ☐ Not available separately

7 - ATF filter

- ☐ Removing and installing
⇒ [page 15](#)
- ☐ Not available separately

8 - Solenoids

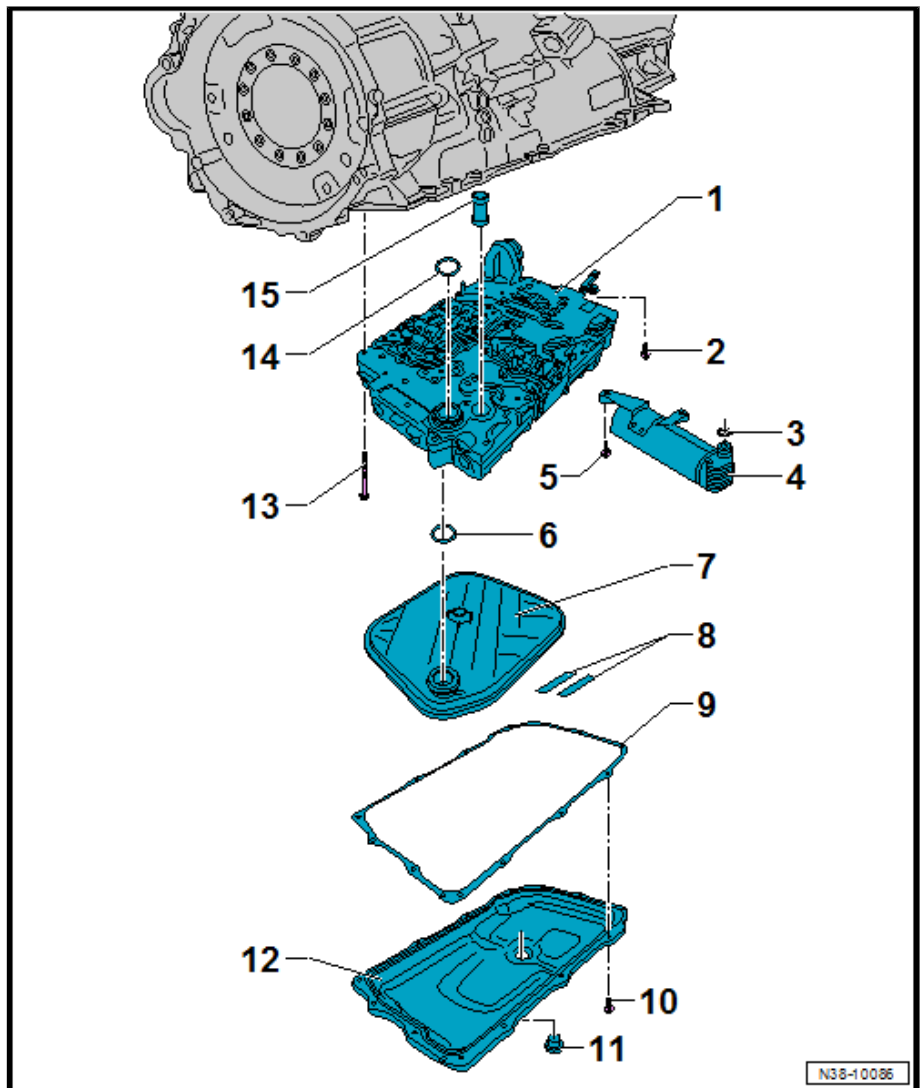
- ☐ Depending on equipment
- ☐ Clean prior to installation
- ☐ Not available separately

9 - Seal

- ☐ Renew after removal
- ☐ Not available separately

10 - Bolts

- ☐ Renew after removal





- ☐ Not available separately
- ☐ Observe tightening sequence ⇒ [page 54](#)
- ☐ Qty. 13
- ☐ 12 Nm

11 - Bolt

- ☐ Not available separately
- ☐ 12 Nm

12 - ATF pan

- ☐ Removing and installing ⇒ [page 52](#)
- ☐ Not available separately

13 - Bolt

- ☐ Renew after removal
- ☐ Not available separately
- ☐ Observe tightening sequence
- ☐ 8 Nm

14 - Seal

- ☐ ATF pump intake
- ☐ Renew after removal

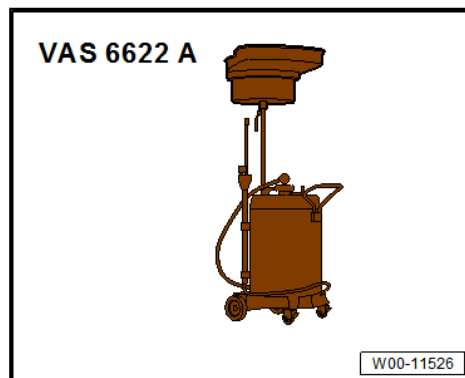
15 - Connecting piece with seal

- ☐ Oil supply for ATF pump
- ☐ Not available separately
- ☐ Renew after removal

1.2 Removing and installing sump

Special tools and workshop equipment required

- ◆ Used oil collector and extractor - VAS 6622A-



- ◆ Torque wrench - V.A.G 1331-





◆ Engine and gearbox jack - VAS 6931-



Removing

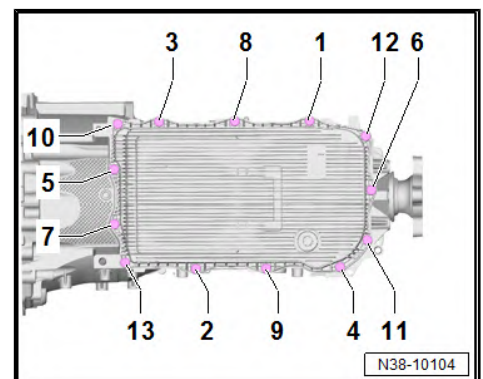
- If fitted, remove underbody guard ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation .
- Remove subframe ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Removing and installing subframe with steering rack .
- Unscrew ATF drain plug and drain ATF ⇒ [page 22](#) .



Note

Support gearbox using engine and gearbox jack - VAS 6931- ensuring that sump can still be removed in the following work procedure.

- Position engine and gearbox jack - VAS 6931- under transfer box.
- Slightly raise gearbox with engine and gearbox jack - VAS 6931- .
- Place used oil collection and extraction unit - VAS 6622A- below gearbox.
- Remove bolts -1 ... 13-.





- Remove sump -1- in direction of -arrow-.

Installing

Install in reverse order of removal, observing the following:

- Clean magnets in sump.



Note

The solenoids must be firmly seated in guide.

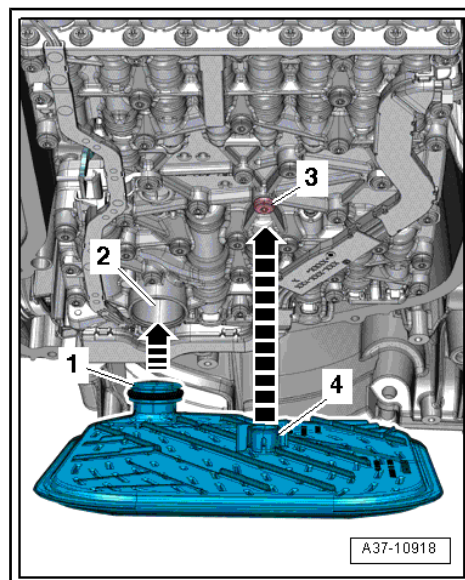
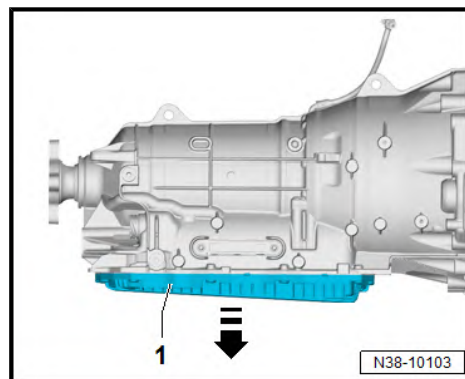
- Thoroughly clean sealing surface.
- Remove remains of old seal.



Note

Ensure proper seating of ATF filter on valve body. Otherwise the ATF sump cannot be fitted properly.

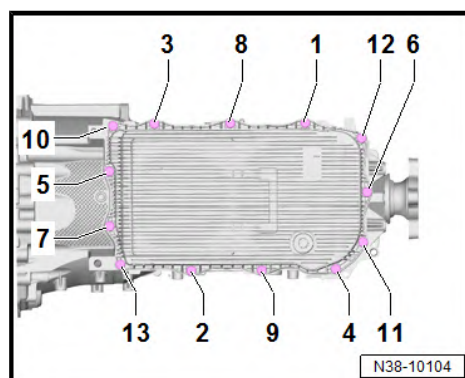
- Collar -1- of ATF filter must be inserted into opening -2- of valve body as far as stop.
- Receiver -4- on rear must engage in bolt -3- of valve body above.



- Tighten bolts in the specified sequence.
- Fill ATF ⇒ [page 22](#) .

Specified torques

- ♦ ⇒ [“1.1 Assembly overview - sump/oil strainer”, page 51](#)
- ♦ Subframe; Assembly overview - subframe ⇒ Running gear, axles, steering; Rep. gr. 40 ; Subframe; Assembly overview - subframe



1.3 Removing and installing hydraulic impulse oil storage with accumulator solenoid - N485-

Removing

- Gearbox is installed.



Caution

Risk of damage to gearbox.

- ◆ *The engine must not be started when there is no ATF in the gearbox and the mechatronic unit is removed.*

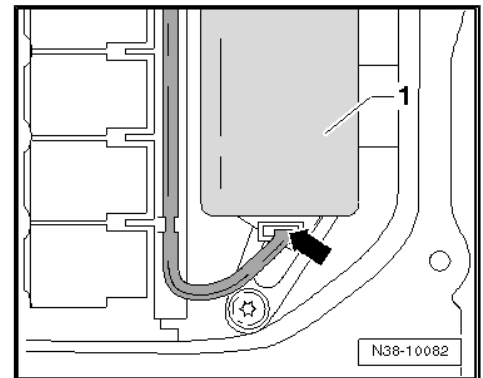
- Shift selector lever to position P.
- Switch off ignition and remove ignition key.
- Remove ATF sump ➔ [page 52](#) .
- Remove ATF filter ➔ [page 15](#) .
- Pull off electrical connector -arrow- on hydraulic pulse accumulator -1-.



Caution

Risk of damaging gearbox control unit (mechatronic unit) through electrostatic discharge.

- ◆ *Before handling electrical connector, the person handling it must "electrostatically discharge" himself/herself. To do so, briefly touch e.g. an earth connection, a radiator in the workshop or a lifting platform with the hand.*
- ◆ *Do not touch contacts in the gearbox connector with hands.*



- Unscrew bolts -arrows- from hydraulic pulse accumulator -1-.
- Remove hydraulic impulse oil storage.

Installing

Install in reverse order of removal, observing the following:



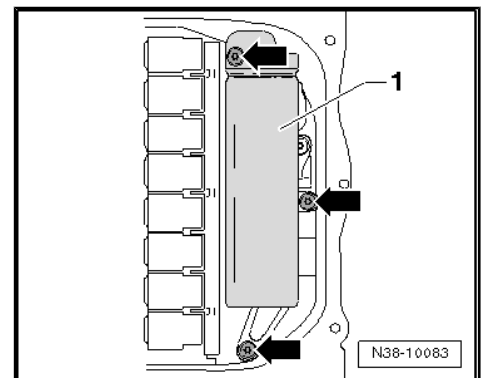
Note

Ensure that connector is properly engaged and secured.

- Insert hydraulic impulse oil storage with new O-ring.
- Install ATF filter ➔ [page 15](#) .
- Install ATF sump ➔ [page 52](#) .
- Fill ATF ➔ [page 22](#) .

Specified torques

- ◆ ➔ ["1.1 Assembly overview - sump/oil strainer", page 51](#)





2 Oil seals in gearbox

⇒ [“2.1 Assembly overview - output shaft flange”, page 56](#)

⇒ [“2.2 Renewing output shaft seal”, page 56](#)

⇒ [“2.3 Removing and installing output shaft flange”, page 58](#)

2.1 Assembly overview - output shaft flange

1 - Nut

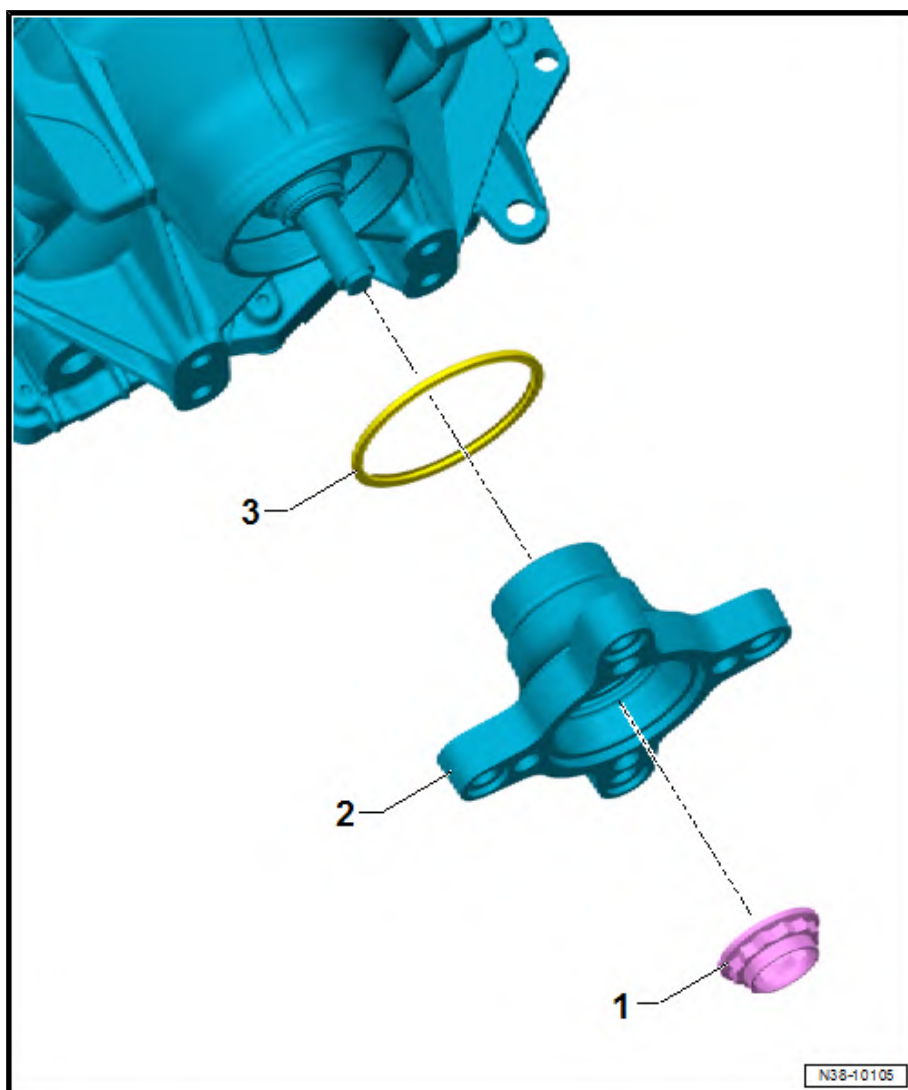
- ☐ Renew after removal
- ☐ Not an individual part
- ☐ 150 Nm

2 - Output shaft flange

- ☐ Not an individual part
- ☐ Removing and installing
⇒ [page 58](#)

3 - Seal

- ☐ Renew after removal
- ☐ Not an individual part
- ☐ Removing and installing
⇒ [page 56](#)

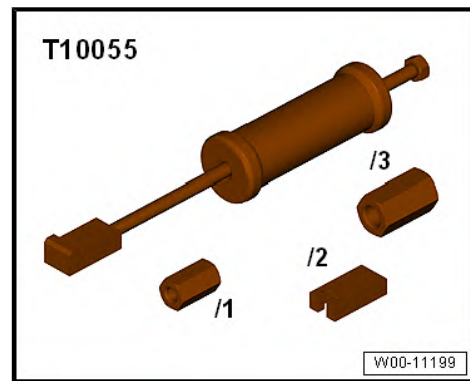


2.2 Renewing output shaft seal

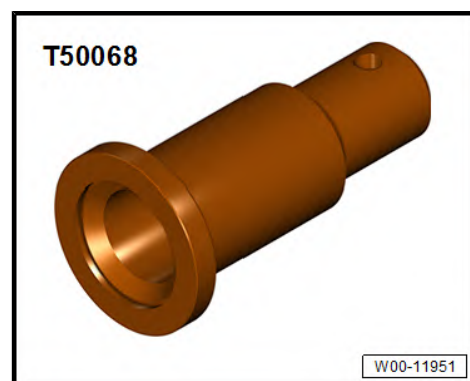
Special tools and workshop equipment required



◆ Puller - T10055-



◆ Thrust piece - T50068-



◆ Self-tapping screw, approx. $\varnothing$ 4 mm

◆ Hand drill

◆ Depth gauge

Removing

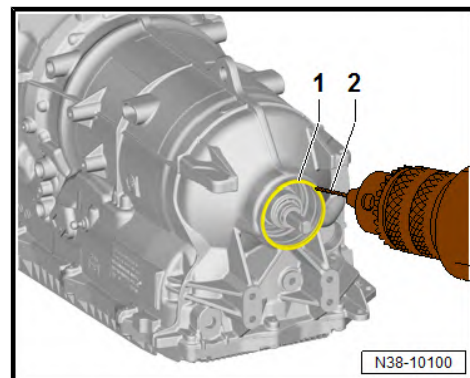
- Remove output shaft flange [⇒ page 58](#) .
- Calculate installation depth of seal using depth gauge and note result.
- Grease drill bit -2-.



Note

Only drill through sheet metal ring -1- (about 3 mm) as the gearbox could otherwise be damaged.

- Using drill bit -2- (2 to 4 mm in diameter), drill through outer sheet metal -1- of seal.





- Screw a self-tapping screw, approx. 4 mm in diameter, into hole drilled in oil seal.
- Pull out seal using puller - T10055- -2- and adapter - T10055/2- -1-.



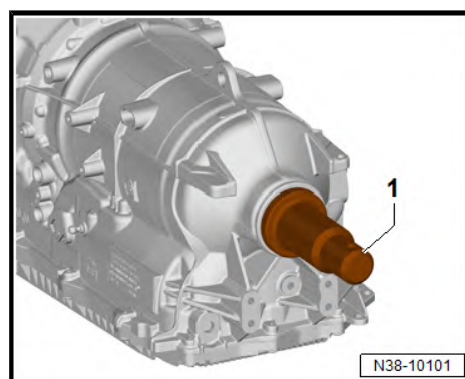
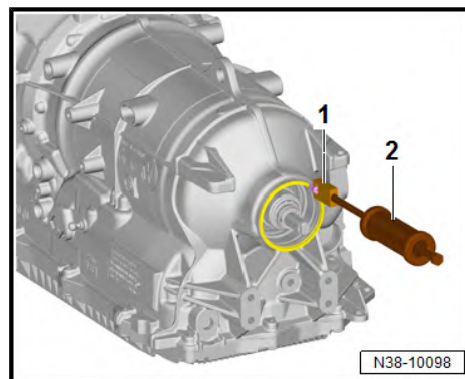
Note

Make sure no metal chips get into gearbox and into opening for drive shaft. Vacuum out metal chips if necessary.

Installing

Install in reverse order of removal, observing the following:

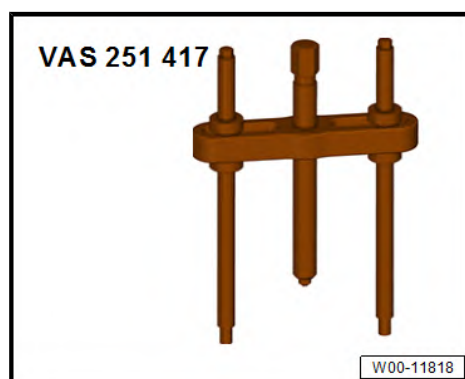
- Press oil seal in straight as far as possible by hand.
- Note installation depth of seal.
- Drive in seal with puller - T10055- -1-. Do not cant oil seal in the process.
- Install output shaft flange ⇒ [page 58](#) .
- Check oil level and top up ⇒ [page 20](#) .



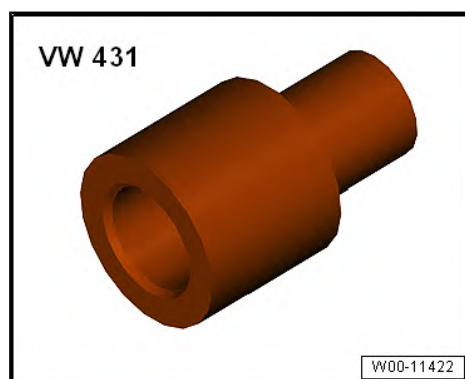
2.3 Removing and installing output shaft flange

Special tools and workshop equipment required

- ♦ Puller, e.g. KUKKO 18-1 - VAS 251 417-



- ♦ Thrust piece - VW 431-





- ◆ Counter-hold tool - 3145-



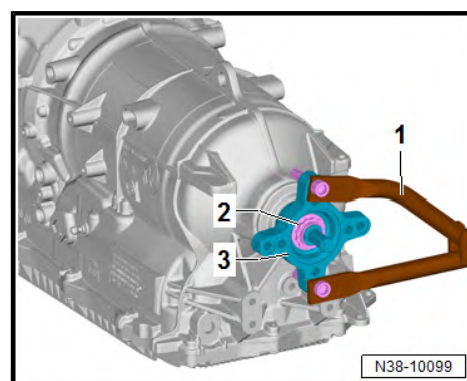
- ◆ Torque wrench - V.A.G 1332-



- ◆ Bolt M12 x 1.5 mm (120 mm in length)

Removing

- Remove propshaft ⇒ Propshaft and rear final drive; Rep. gr. 39 ; Propshaft; Removing and installing propshaft .
- Secure counterhold - 3145- -1- to output flange -3-.
- Unscrew nut -2-. Counterhold using counterhold - 3145- -1- when doing this.



- Mount puller, e.g. KUKKO 18-1 - VAS 251 417- -1- with thrust piece - VW 431- .
- Pull out output flange -2- using puller, e.g. KUKKO 18-1 - VAS 251 417- -1- from gearbox -3-.

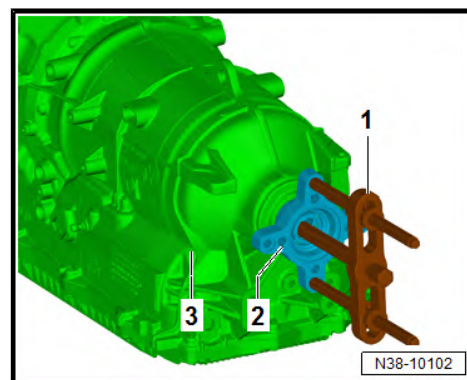
Installing

Install in reverse order of removal, observing the following:

- Install propshaft ⇒ Propshaft and rear final drive; Rep. gr. 39 ; Propshaft; Removing and installing propshaft .

Specified torques

- ◆ ⇒ ["2.1 Assembly overview - output shaft flange", page 56](#)





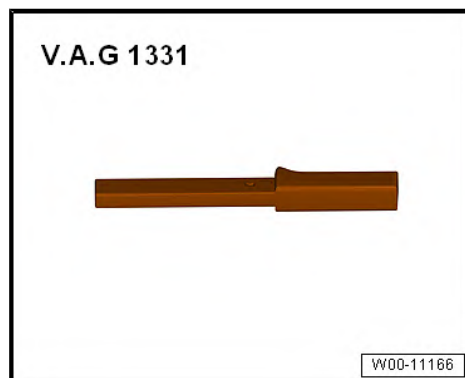
3 Valve body

⇒ ["3.1 Removing and installing valve body", page 60](#)

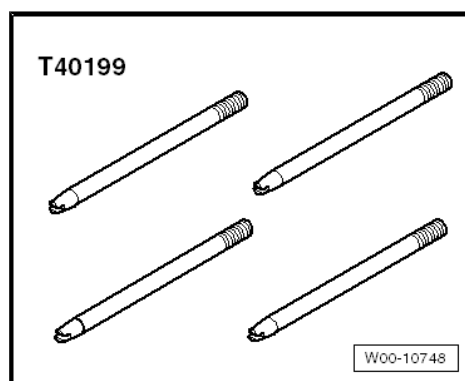
3.1 Removing and installing valve body

Special tools and workshop equipment required

◆ Torque wrench - V.A.G 1331-



◆ Guide pin - T40199-



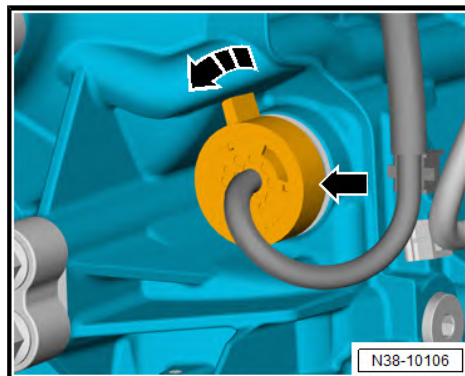
Removing



WARNING

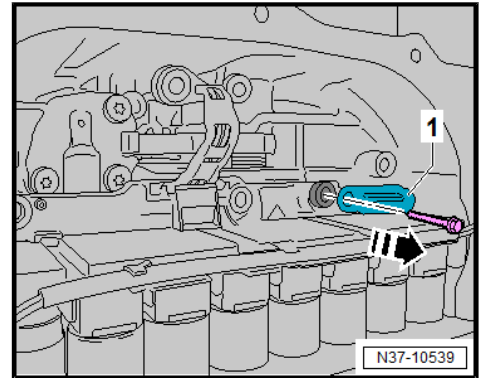
Do not run engine or tow vehicle with pan removed or when there is no ATF in gearbox.

- Shift selector lever to position P.
- Switch off ignition.
- Remove ATF sump ⇒ [page 52](#) .
- Remove ATF filter ⇒ [page 15](#) .
- Disconnect connector -arrow- by turning in direction of -arrow-.





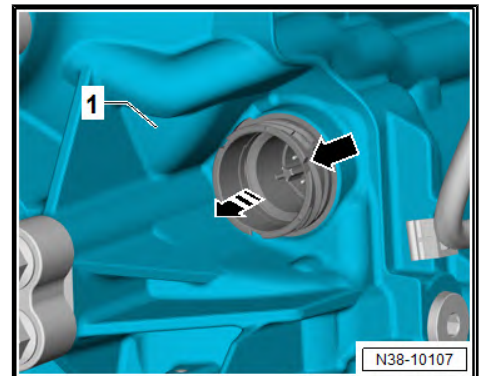
- Unscrew bolt and pull catch -1- of connector in direction of -arrow-.



- Pull out connector housing -arrow- from gearbox -1- in direction of -arrow-.

i Note

- ◆ Only the marked bolts may be loosened.
- ◆ Loosening other bolts may affect the operation of the valve body, or the valve body could come apart.



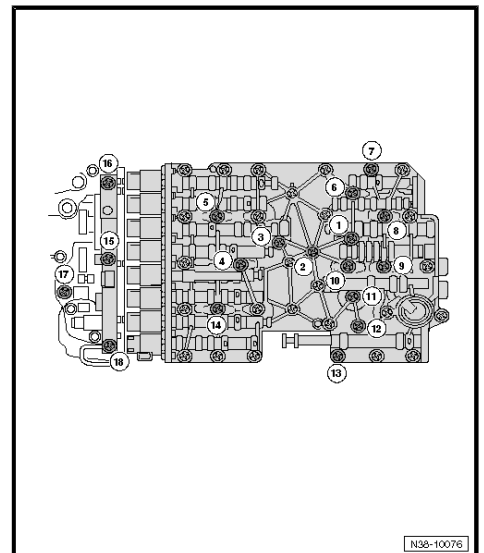
i Note

- ◆ Bolt -16- has already been removed during removal of connector housing.
- ◆ Do not yet remove bolt -1-.

- Unscrew bolts in sequence -18 to 2-.
- Pull out gearbox output speed sender - G195- after bolt -17- has been loosened, in order to prevent the sender from being damaged.
- Screw in guide pins - T40199- hand-tight into holes -2, 5, 13, 14-.
- Unscrew bolt -1-.

i Note

Only lay down the valve body with the bolt head side facing downwards.





- Remove valve body downwards.

- 1 - Electronic module with sensors
- 2 - Electronic gearbox control
- 3 - Hydraulic valve body
- 4 - Solenoid valves
- 5 - Electrical connector

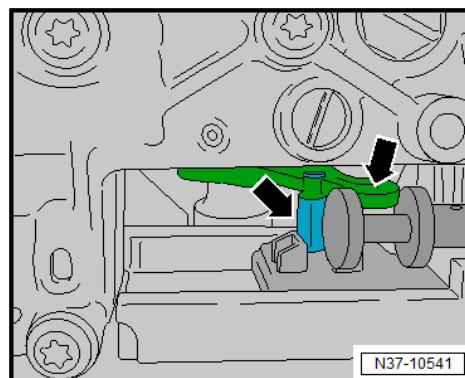
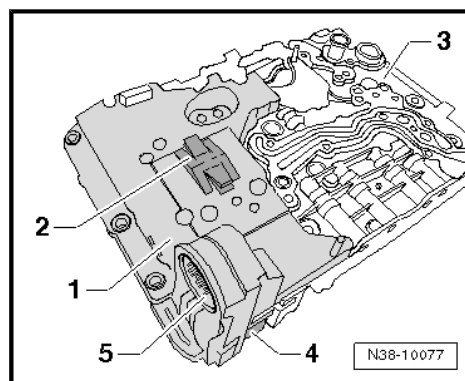
Installing

Install in reverse order of removal, observing the following:

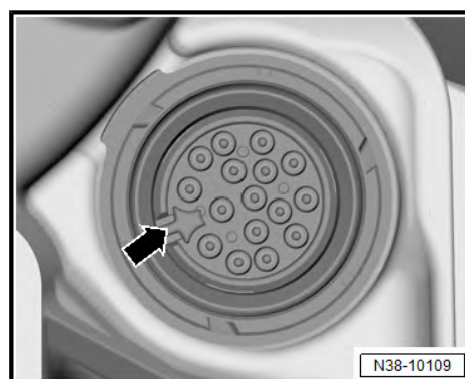


Note

- ◆ *Only use lint-free cloths.*
- ◆ *Always renew a valve body which is fouled or defective.*
- The following work step must be carried out with the aid of another mechanic.
- Gearbox selector lever must be seated in groove of slide block and selector slide -arrows-.
- Screw in bolts -1, 3, 4- by hand until bolt heads make contact.
- Unscrew 4 guide pins - T40199- .



- Install new connector housing (prior to installing bolt -16-). Observe locking lugs -arrows- when doing this.
- Then, lock securing clip.
- Do not tighten bolt -16- until after the securing clip has been properly locked.
- Tighten bolts for mechatronic unit in prescribed sequence.

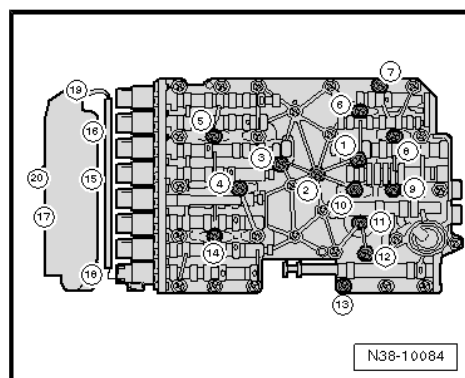


Tightening sequence for mechatronic unit

- Install ATF sump ⇒ [page 52](#) .
- Install ATF filter ⇒ [page 15](#) .
- Fill ATF ⇒ [page 22](#) .
- Adapt mechatronic unit using ⇒ Vehicle diagnostic tester.

Specified torques

- ◆ ⇒ ["1.1 Assembly overview - sump/oil strainer", page 51](#)





4 Gearbox control system

⇒ ["4.1 Removing and installing automatic gearbox control unit J217", page 63](#)

4.1 Removing and installing automatic gearbox control unit - J217-

Removing

- Automatic gearbox control unit - J217- ⇒ Current flow diagrams, Electrical fault finding and Fitting locations.

Vehicles with second battery

- Battery; Remove battery tray ⇒ Electrical system; Rep. gr. 27 ; Battery; Removing and installing battery tray .

Continued for all vehicles

- Disconnect electrical connector -2-.
- Detach automatic gearbox control unit - J217- -1- from holder.

Installing

Install in reverse order of removal.

